What's in a Name? Disagreement about Discrimination in Tokyo's Rental Housing Market*

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Abstract

We study beliefs about market access in Tokyo's rental housing sector. Despite growing evidence that beliefs about discrimination have direct economic consequences, independently of realised discriminatory behaviour, little is known about how majority and minority group perceive the same minority disadvantage at the group and individual levels. We use a name-based survey experiment in Tokyo where native Japanese and foreign-born residents evaluate whether otherwise identical applicants would receive a viewing invitation. Majority respondents perceive lower access probabilities for minority applicants than minority respondents do. Minority respondents, in turn, perceive higher access probabilities for themselves than for minority applicants generally. Both groups perceive minority disadvantage, but minority respondents perceive less of it, a pattern we term the Group-Group Discrimination Discrepancy (GGDD). They also perceive less disadvantage for themselves than for their group, consistent with the established Personal-Group Discrimination Discrepancy (PGDD).

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1 Introduction

Japan is an unusual setting in which to study perceived market-access discrimination. Foreign residents remain a small share of the population relative to other high-income economies, and comprehensive statutory protection against nationality-based discrimination in private housing is absent: unlike the United Kingdom's *Equality Act 2010* or the European Union's *Racial Equality Directive* (2000/43/EC), no comparable statute extends to Japan's rental sector (subsection 2.2).[1] Yet this is also a moment in which the state has moved, cautiously and without officially calling it an immigration policy, to actively recruit more foreign labour. The 2018 reform created two new visa categories, the Specified Skilled Worker Type 1 and 2, explicitly recognising the need for lower-qualified labour (Chiavacci, 2025), even as former Prime Minister Shinzo Abe stated during a Diet session on 29 October 2018 that the government had no intention to adopt a so-called immigration policy.[2] This combination—a small, legally under-protected minority that the state is simultaneously trying to grow—raises a question existing discrimination research does not directly address: do the country's likely future hosts and its likely future arrivals share the same beliefs about how open the market actually is to them, or do they systematically disagree?

A substantial literature already shows that members of a disadvantaged group do not evaluate their own prospects and their group's prospects identically: minority respondents typically report less personal disadvantage than the disadvantage they attribute to their broader group, a pattern termed the *Personal-Group Discrimination Discrepancy* (PGDD) and documented across labour, income, and—more sparsely—housing contexts (Crosby, 1982; Taylor et al., 1990, 1996; Dion, 2001; McAvay and Safi, 2025; Martiniello, 2025). The classic pattern is summarised by Taylor et al. (1996) as "perceiving my group, but not myself," and housing-specific evidence remains scarce, confined mainly to Canada (Dion, 2001) and, more recently, Europe (Martiniello, 2025). What this literature has not established is whether majority and minority observers agree on the magnitude of disadvantage facing the minority group in the first place: PGDD compares a minority respondent's beliefs about themselves against their beliefs about their own group, but says nothing about whether an outside majority observer, asked to evaluate that same group, would arrive at the same assessment. We answer this question for Tokyo's rental market and, in doing so, make two contributions. Conceptually, we introduce and measure this previously unisolated belief gap: the *Group-Group Discrimination Discrepancy* (GGDD), the extent to which majority and minority observers disagree about the very same minority group's disadvantage, complementing PGDD as a second, distinct axis along which perceived discrimination can diverge. Empirically, we provide, to our knowledge, the first evidence on this question for Japan at a moment when the disjunction between policy ambition and legal protection makes the distribution of such beliefs directly consequential: if the state is trying to expand a workforce it does not yet fully protect from discrimination, whether majority and minority residents even agree on how large that protection gap is becomes a first-order empirical question rather than a peripheral one.

Beliefs about discrimination are economically consequential independently of realised discriminatory behaviour. Early work in economics recognised the impact of perceived discrimina-

[1] *The Economist* recently argued that Japan's perceived "foreigner problem" is in fact the opposite: rather than hosting too many foreigners, Japan faces a shortage of foreign workers at a time of rapid population ageing. The article further highlights substantial regional differences. While over-tourism has fuelled concerns about foreigners in destinations such as Kyoto, many ageing rural communities increasingly rely on foreign workers to sustain local industries and even traditional festivals. See *The Economist*, "Does Japan have a 'foreigner problem'? Yes—but it is not what populist politicians say it is", January 8, 2026, https://www.economist.com/leaders/2026/01/08/does-japan-have-a-foreigner-problem.

[2] https://mainichi.jp/english/articles/20181029/p2a/00m/0na/032000c, last accessed in May 2026.

tion learned from past experience on subsequent economic behaviour, integrating neoclassical labour-supply theory with cognitive dissonance (Goldsmith, 2002). More recent research shows that anticipated discrimination itself triggers strategic adaptation, regardless of any observable discriminatory behaviour: minority candidates may respond by offering higher prices to compensate an implicit access premium (Zussman, 2013), greater compensatory investment (Dickerson, 2024), hiding names and other identities (Charness, 2020; Kudashvili and Lergetporer, 2020; Aksoy, 2023), or not applying altogether. On the majority side, discriminatory behaviour is sensitive not only to private preferences but also to perceived social norms regarding the acceptability of discrimination (Barr et al., 2018). In the housing market specifically, anticipated rejection may influence search intensity, application strategies, neighbourhood choice, and residential sorting, and Boeri et al. (2015) show that housing-market discrimination may affect immigrant integration beyond housing itself by shaping residential concentration and employment outcomes. Beliefs about minority market access therefore shape the behaviour of both minority and majority participants across the market, making the distribution of those beliefs an important object of study in its own right.

A separate line of stereotype research shows that beliefs about the same social group may overlap but systematically differ depending on the group the evaluator belongs to. Katz and Braly (1933) elicits ethnic trait stereotypes among Princeton students. Building on the Katz-Braly tradition, Maykovich (1972) studies the reciprocity of racial stereotypes by comparing White, Black, and Japanese American respondents' perceptions of each racial group, while Judd and Park (1993) compares in-group and out-group beliefs about the same target group, with a particular focus on stereotype accuracy—a concern structurally related to, but conceptually distinct from, GGDD, since our fictitious applicant profiles admit no independently verifiable criterion against which either belief could be scored as more accurate. Clark and Pearson Jr (1982) revisits the Katz-Braly tradition and documents substantial overlaps, alongside systematic discrepancies, across self-rating, auto-stereotype, and hetero-stereotype among Black and White college students. Despite this line of work, we are not aware of evidence on whether majority and minority share the same beliefs about the market access of a minority group specifically. Eliciting hetero-stereotype directly is also challenging, in part because direct measures of majority prejudice are vulnerable to social-desirability pressure, by which respondents adjust their answers toward what is socially acceptable once they recognise a question is testing them for prejudice.

We find that both native Japanese and foreign-born respondents expect non-Japanese applicants to face lower viewing probabilities than an otherwise identical Japanese applicant, with the penalty graded by profile rather than binary. Foreign-born respondents rate their own prospects more favourably than those of their broader nationality group—consistent with PGDD—while native Japanese respondents perceive larger penalties for the same non-Japanese applicants than foreign-born respondents do, evidence of GGDD. Supporting analyses show that prior experiences of everyday discrimination are associated with more pessimistic personal expectations, while differences in Japanese-language proficiency provide suggestive, rather than causal, evidence that access to market information may help explain variation in group-level perceptions. A complementary platform exercise further establishes that nationality-related screening language remains publicly visible on major rental platforms, without estimating its prevalence across the full universe of listings.

The remainder of the paper is organised as follows: section 2 introduces Tokyo's rental market, the applicable legal framework, and the broader immigration context; section 3 describes the survey design and experimental methodology; section 4 presents the empirical results and the platform-visibility exercise; and section 5 discusses results and derives policy recommendations.

Finally, section 6 concludes. The Appendix reports supplementary materials.

2 Legal and Institutional Context

This section begins with the structure of Tokyo's rental market and the role of private tenant screening. It then sets out the comparatively weak statutory constraints on nationality-based exclusion before placing these market institutions within Japan's broader immigration regime and ethnic hierarchy. This ordering distinguishes the immediate housing-market setting from the wider social and political context used to interpret the results.

2.1 Tenant-Protective Property Law and the Spatial Structure of Tokyo

Tokyo's contemporary rental housing market is shaped by a historically entrenched system of strong tenant protections and long-term reliance on rental housing. Unlike housing systems in which renting primarily serves as a temporary stage before homeownership, renting constitutes a central and often long-term tenure form in Tokyo. This makes access to rental housing particularly important for social and economic integration, especially for foreign-born residents.

The institutional structure of Tokyo's rental market reflects a longer historical development involving the evolution of tenant protections, the historical organisation of urban landholding, and postwar patterns of rental dependence. In particular, Japan's strong tenant-protection regime, including automatic lease renewal and the "just cause" requirement, has historically encouraged relatively stable and long-term tenancy relations while also contributing to risk-averse screening behaviour among landlords. Additional historical background on the development of Tokyo's rental housing market is provided in Appendix A.

Despite the introduction of fixed-term leases in 2000, much of Tokyo's rental market remains structurally oriented toward relatively stable occupancy rather than short-term tenant turnover. Combined with comparatively low homeownership rates in Tokyo, this institutional structure increases the importance of equitable access to rental housing.

Existing research documents persistent socio-economic and spatial inequalities within Tokyo's housing market. Although segregation levels are generally lower than in many Western metropolitan areas, disparities remain visible across occupational, income, and ethnic dimensions (Masaya, 2021; Fielding, 2004; Fujita and Hill, 2012). Foreign-born residents are concentrated in specific wards such as Shinjuku and Edogawa, reflecting both labour-market sorting and historical migration networks.[3]

Tokyo's rental market therefore provides an important setting in which to study perceived discrimination. Access to rental housing is frequently mediated through private landlords and real-estate agents, creating substantial scope for unequal treatment at the screening stage. Existing studies further suggest that social perceptions and cultural proximity continue to shape attitudes toward foreign-born residents in Japan (Davison and Peng, 2021; Shipper, 2016).

2.2 Weak Anti-discrimination Law

Existing evidence suggests that exclusionary behaviour in Japan's housing market occurs at multiple stages of the rental process. Correspondence evidence documents differential responses to otherwise comparable Japanese and foreign applicants at the initial inquiry stage, while repeated nationwide surveys commissioned by the Japanese Ministry of Justice report that

[3] Immigration Services Agency (2024) data indicate that Shinjuku has the largest foreign-born resident population among Tokyo's 23 wards.

foreign residents frequently experience difficulties securing housing (Sugasawa and Harano, 2023). More recent evidence shows that this disadvantage is stronger among agencies with post-tenancy management responsibilities, although it also persists among brokerage-only agencies (Sugasawa and Harano, 2026). Using administrative records from moving-in inspections and subsequent rent payments in the Tokyo metropolitan area, Suzuki et al. (2022) further show that foreign applicants and other tenants classified as atypical are less likely to receive approval even after accounting for their predicted probability of rent arrears and observable payment ability. The evidence therefore suggests that unequal access is not confined to initial contact and cannot be explained solely by observable differences in payment risk.

Article 14 of the Japanese Constitution prohibits discrimination on several grounds, including race, creed, sex, social status, and family origin. Japan also became a party to the International Convention on the Elimination of All Forms of Racial Discrimination (ICERD) in 1995, which entered into force domestically in 1996. In the housing context, Japanese courts have recognised certain forms of nationality-based differential treatment as unlawful under tort law and the principle of good faith in the Japanese Civil Code, particularly where landlords or real-estate agents refused to conclude tenancy agreements with non-Japanese applicants.[4] More recently, the Tokyo District Court awarded damages against a real-estate agency that expressed unwillingness to rent to persons from a particular country.[5]

However, unlike the United Kingdom and European Union member states, Japan does not have statutory anti-discrimination provisions explicitly regulating differential treatment in the housing sector, nor an independent national human rights or equality body with monitoring/enforcement powers. In the United Kingdom, the *Equality Act 2010* expressly prohibits race discrimination in relation to the disposal and management of premises in the housing sector.[6] At the European Union level, the *Racial Equality Directive* (2000/43/EC) requires Member States to prohibit discrimination in access to and supply of goods and services available to the public, including housing, both in the public and private sector. The Directive further requires Member States to designate bodies for promotion of equal treatment and establish effective, proportionate, and dissuasive sanctions for infringements of anti-discrimination law.[7] All Member States have now transposed this Directive to their national laws.[8]

A lack of statutory anti-discrimination provisions is consistent with the continued public visibility of nationality-based screening language within the Japanese housing sector, examined empirically in subsection 4.4. This contrasts with the position in jurisdictions such as the United Kingdom, the European Union, and the United States.[9] Taking the United Kingdom as an example, race, including nationality, is a protected characteristic under anti-discrimination law,[10] and the Equality and Human Rights Commission's Code of Practice treats advertisements expressing preferences or restrictions based on race or nationality amount to direct discrimination.[11] Accordingly, such advertisements are themselves unlawful, irrespective of whether

[4]Osaka High Court, 5 October 2006; Osaka High Court, 22 March 2007; Kyoto District Court, 2 October 2007.

[5]Tokyo District Court, 9 October 2019.

[6]Equality Act 2010, ss4, 9(1)(b), 13, 19, 26, 27 and 33-35.

[7]Art 3(1)(h), 13, 15.

[8]European Commission (2007), "The Race Equality Directive," available at `https://ec.europa.eu/commission/presscorner/detail/en/memo_07_257` (accessed 17 May 2026).

[9]In the United States, the Fair Housing Act 42 USC ğ3604(c) explicitly prohibits discriminatory housing advertisements based on race or national origin.

[10]Equality Act 2010 s9; as to nationality, see s9(1)(b); *Mandla v Dowell Lee* [1983] AC 548; *Ealing LBC v CRE* [1972] AC 342.

[11]Equality Act 2010 s13; Equality and Human Rights Commission, 'Guidance: Services, Public Functions and Associations: Code of Practice' (published 1 January 2011) 4.31 states 'If a service provider advertises

any individual applicant can demonstrate discriminatory treatment in a particular transaction. Unlike indirect discrimination, which may be justified as a proportionate means of achieving a legitimate aim, direct race discrimination generally admits of no comparative justification defence.[12] Under UK equality law, deliberately segregating service users on racial grounds is treated as less favourable treatment.[13] From this perspective, the existence of housing providers or property listings specifically marketed as suitable for foreign national would not necessarily resolve concerns about discrimination if their practical effect is to channel foreign applicants into separate segments of the housing market rather than ensuring equal access the market as a whole.

The United Nations' *Committee on the Elimination of Racial Discrimination* repeatedly expressed concerns regarding discriminatory treatment by private actors in Japan, particularly discrimination in housing. The Committee has also criticised the absence of statutory provisions explicitly prohibiting racial discrimination, the lack of effective remedies, and weak institutional enforcement mechanisms.[14] In response, the Japanese government argued that the *Human Rights Bureau* within the *Ministry of Justice* engages in protecting human rights.[15] While the Bureau, originally established in 1948, explicitly recognises nationality-based exclusion in the housing market as a human-rights issue, it lacks institutional independence from the government and does not possess binding enforcement powers.

2.3 Hate Speech Regulation without Criminal Sanction

Because perceptions of discrimination are often influenced not only by direct experiences but also by publicly expressed exclusionary attitudes, the Japanese legal and regulatory treatment of discriminatory speech is also relevant to the present study. Japan maintains reservations to Article 4(b) and (c) of CERD, which require the criminalisation of certain forms of hate speech and racist organisations, citing concerns regarding freedom of expression. The Committee has

that in offering a service they will treat applicants less favourably because of a protected characteristic, this would amount to direct discrimination'. Accessible at `https://www.equalityhumanrights.com/equality/equality-act-2010/codes-practice/services-public-functions-and-associations-code-0` In any case, under Equality Act 2006 s24A, the Comission has power to apply to the court for an injunction to deal with the publication of an advertisement suggesting that a service provider would discriminate (see the Code 14.61).

[12] cf. Equality Act 2010 s19(2)(d).

[13] Equality Act 2010 s 13(5).

[14] See Committee on the Elimination of Racial Discrimination, *Concluding Observations of the Committee on the Elimination of Racial Discrimination: Japan*, CERD/C/304/Add.114 (2001), available at: `https://docstore.ohchr.org/SelfServices/FilesHandler.ashx?enc=lyJrhoSIHCA2RTKXgko4JqwonaM2kkcNfU6wkv17W5MYLT%2Bv2S5EYH9NmA5629puv7EaR830WENKIUUL4jc1H5tQi2u1ZbAKryD4rPoRvU8%3D`; Committee on the Elimination of Racial Discrimination, *Concluding Observations of the Committee on the Elimination of Racial Discrimination: Japan*, CERD/C/JPN/CO/3-6 (2010), available at: `https://docstore.ohchr.org/SelfServices/FilesHandler.ashx?enc=izUhol7ZxPbJFSoWm%2FljbKdvAC70A0joqH39rxwD%2FsjYQ8OFqNjARt%2BgGI6FwhLsR%2BHOf%2BiMsoHQxD9AuibXWA%3D%3D`; Committee on the Elimination of Racial Discrimination, *Concluding Observations on the Combined Seventh to Ninth Periodic Reports of Japan*, CERD/C/JPN/CO/7-9 (2014), available at: `https://docstore.ohchr.org/SelfServices/FilesHandler.ashx?enc=eH37LT%2F%2BEGaGZEmrzoZ0Bddj8Jx7lStPMHnyoeIwYtHVqaweoS%2F%2FM7RevYJU%2BKbVsT5VOFJoEqW2Avpbo%2FLTBw%3D%3D`; Committee on the Elimination of Racial Discrimination, *Concluding Observations on the Combined Tenth and Eleventh Periodic Reports of Japan*, CERD/C/JPN/CO/10-11 (2018), available at: `https://docstore.ohchr.org/SelfServices/FilesHandler.ashx?enc=cSiLKueFO8pNvNu3RBjBpHu596d6AmEKyPsxtP5iedXv%2FlH9b8GO3JWV4wBKUZ2uVwJCQ%2FBBeEPk4BWzF3NPsWoSunJWsZ8oB%2F6pY7tcdcw%3D`.

[15] Committee on the Elimination of Racial Discrimination, *Information Received from Japan on Follow-up to the Concluding Observations*, CERD/C/JPN/CO/10-11/Add.1 (2019), available at: `https://docstore.ohchr.org/SelfServices/FilesHandler.ashx?enc=%2BTnmP2Rfpf5TFxOTwIkDWOrxS6gOv7bNqxQ1hx%2Fa6i%2FOTwuUXfnDt1habGcxZanbpGTH9jbsgcHLfAUYrqLW1NB%2Bon%2Fg3nSq7UOSnjG85zc%3D`

repeatedly expressed concerns regarding racist and xenophobic speech in Japan.[16]

In response to growing domestic and international criticism, Japan enacted the *Hate Speech Act 2016*.[17] However, the Act has a limited scope and lacks enforcement structure. The definition of prohibited conduct is relatively narrow, requiring speech or behaviour intended to promote or induce discriminatory sentiments or discriminatory behaviour (Art 2). Moreover, the Act does not impose any sanctions, instead providing only that 'citizens of Japan shall endeavour to deepen their understanding of the need to eliminate unfair discriminatory speech or behaviour' (Art 3). While Japanese courts have, in some cases, awarded damages against hate speech both offline and online,[18] the CERD Committee has continued to express concerns regarding the absence of effective anti-hate speech enforcement mechanisms in Japan.[19]

Taken together, Japan's legal and institutional framework provides comparatively limited constraints on nationality-based exclusion in private rental markets. While courts have occasionally recognised particularly explicit forms of discrimination as unlawful, the absence of comprehensive anti-discrimination legislation and effective enforcement mechanisms leaves substantial discretion to landlords and real-estate agents in screening prospective tenants.

However, the Japanese government maintains that discrimination against non-Japanese residents was not sufficiently severe to require comprehensive anti-discrimination legislation or criminal sanctions. It stated that 'We do not recognize that the present situation of Japan is one in which discriminative act cannot be effectively restrained by the existing legal system...', thereby maintaining that the existing Japanese legal framework was sufficient to address discriminatory conduct without further legislative intervention.[20] The empirical analysis therefore examines separately whether nationality-related screening signals remain publicly observable in Tokyo's contemporary rental market.

2.4 Broader Immigration Context and Ethnic Hierarchies in Japan

For decades, Japan denied the need for an immigration policy, instead implementing a system of de facto labour importation through side and back door entry while limiting pathways to long-term settlement and integration for foreign nationals (Vermillo Herman, 2021). In official discourse, the Japanese government has continued to reject the notion of 'immigration,' treating immigration as implying permanent settlement and integration (Hasegawa, 2023). Similar dynamics have been observed more broadly across East Asian migration regimes, where foreign workers are frequently incorporated as economically necessary labour while remaining socially and politically temporary populations (Bell and Piper, 2005). In such systems, migrants are often treated as labour market participants rather than prospective long-term members of the national community.

[16] See note 7.

[17] Act of the Promotion of Efforts to Eliminate Unfair Discriminatory Speech and Behaviour against Persons Originating from Outside Japan (2016).

[18] For instance, the Kyoto District Court (7 October 2013) awarded damages against an extremist anti-Korean organisation whose members repeatedly used loudspeakers near a Korean school to disseminate discriminatory and derogatory speech targeting ethnic Koreans. More recently, the Tokyo High Court (12 May 2021) awarded damages that were relatively high by Japanese tort law standards against a blogger, who had posted discriminatory and derogatory comments against ethnic Koreans.

[19] See *Concluding Observations on the Combined Tenth and Eleventh Periodic Reports of Japan* (2018), in note 11.

[20] Ministry of Foreign Affairs of Japan, *Comments of the Japanese Government on the Concluding Observations Adopted by the Committee on the Elimination of Racial Discrimination on March 20, 2000, Regarding Initial and Second Periodic Report of the Japanese Government* (2000), available at: `https://www.mofa.go.jp/policy/human/comment0110.html`.

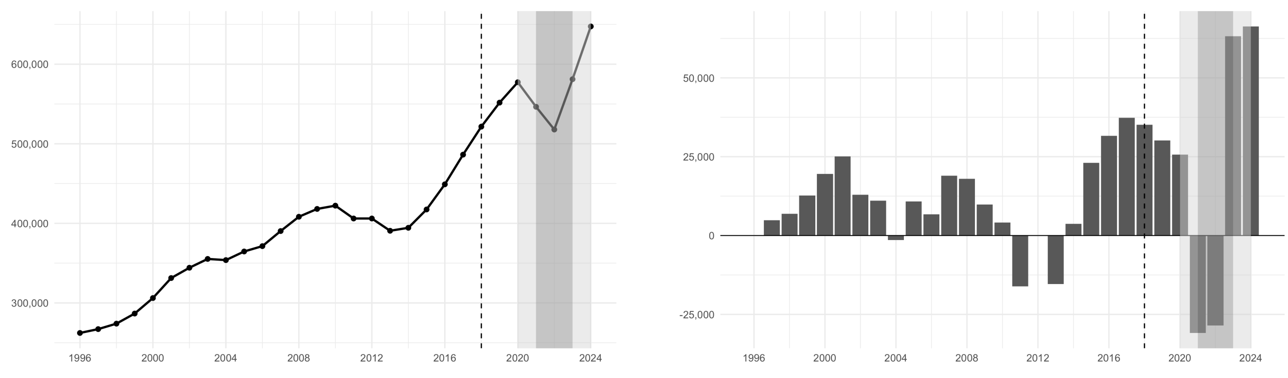
This distinction between labour admission and immigration resembles the broader logic of guestworker regimes, which admit foreign workers as a temporary and functionally necessary labour force while restricting claims to settlement, family life, and political membership (Surak, 2013). While the 2018 reform introduced two new visa categories — Specified Skilled Worker (SSW) Type 1 and 2 — explicitly recognising the need for lower-qualified labour (Chiavacci, 2025), former Prime Minister Shinzo Abe nevertheless stated during a Diet session on 29 October 2018[21] that 'the government has no intention to adopt a so-called immigration policy.' Despite this official framing, the reform marked a substantial shift in Japan's migration regime. Chiavacci (2025) even characterises the reform as "a dam break in Japan's immigration policy," reflecting the disruptive nature of the institutional transition toward greater reliance on foreign labour. This transition increases the importance of understanding how foreign-born residents are perceived within key markets such as housing, where access depends heavily on private screening decisions.

2.5 Ethnic Hierarchies and Differential Integration

The 2018 reforms were primarily driven by demographic pressures, labour shortages, and coordinated business lobbying rather than a broader shift toward multicultural integration (Vermillo Herman, 2021). Despite institutional reforms, immigration policy in Japan continues to reflect ethnonationalist frameworks and differentiated perceptions of cultural compatibility (Shipper, 2016). Existing research suggests that immigrants are incorporated into a relatively clear ethnic hierarchy, in which co-ethnic and culturally proximate groups are often treated more favourably than other migrant populations (Tsuda, 1998). In the Japanese context, anti-immigrant attitudes are further found to be strongly shaped by concerns regarding identity and cultural homogeneity (Davison and Peng, 2021).

Figure 1 illustrates the increase in Tokyo's foreign resident population following the 2018 immigration reform, despite a temporary slowdown during the COVID-19 period.

Figure 1: Foreign Residents in Tokyo, 1996–2024



(a) Total Foreign Residents in Tokyo

(b) Annual Change in Foreign Residents

Notes: The dashed vertical line marks the *2018 Immigration Control and Refugee Recognition Act* reform. Shaded areas indicate the COVID-19 period. The darker shaded region corresponds to years fully affected by pandemic-related restrictions, while lighter shaded regions correspond to partially affected years. Data are from the *Tokyo Statistical Yearbook*.

Corresponding figures for individual nationality groups are shown in Figure 7 in the Appendix.

[21] https://mainichi.jp/english/articles/20181029/p2a/00m/0na/032000c, last accessed in May 2026.

3 Survey Design

This section describes the design of a survey experiment intended to elicit beliefs about rental-market access directly, rather than to infer such beliefs indirectly from realised landlord or agent behaviour, as correspondence and audit studies typically do. This follows from the premise, set out in section 1, that perceived discrimination is not merely a report on an independently existing market friction but is itself constitutive of that friction; eliciting the belief directly is accordingly the most direct available route to observing the friction, rather than a second-best substitute for measuring realised outcomes.

3.1 Technical Implementation and Quality Assurance

We conducted a cross-sectional online survey between April and May 2025 using the Japanese panel provider *GMO Research*. The survey targeted residents of the Greater Tokyo area and used quota sampling to recruit approximately 800 Japanese nationals and 200 foreign-born residents. Foreign-born respondents were oversampled to ensure sufficient power for subgroup analyses.

The questionnaire was implemented in Qualtrics and included standard data-quality safeguards, including minimum completion-time thresholds, response-completeness checks, location validation, and duplicate-response prevention. After applying these filters, a small number of observations was excluded due to incomplete or inconsistent responses. Further implementation details are reported in Appendix B. Ethical approval was obtained from the Department of Land Economy, and informed consent was secured from all participants prior to participation.

3.2 Development of Fictitious Rental Applicants

Applicant profiles were constructed using *Japanese Immigration Agency* (2024) data to reflect the composition of migrant inflows following the 2018 immigration reforms. To isolate perceived discrimination, profiles differed only in ethnically identifiable names, while all other applicant characteristics were held constant. Applicant names were pre-tested for ethnic recognisability and standardised using Japanese script conventions to preserve comparability across profiles. Table 1 reports the final applicant profiles.

Table 1: Applicant Profiles

Name	Ethnicity	Native Translation	Katakana Characters
Taro Yamamoto	Japanese	山本太郎	山本太郎
Andi Surya Purta	Indonesian	Andi Surya Purta	アンディ・スルヤ・プトラ
Wei Chang	Chinese	张伟	ウェイ・チャン
Thomas Laurent	Western	Thomas Laurent	トーマ・ローラン
James Takeda	Western-Japanese	James Takeda	ジェームズ・武田
Nguyễn Văn Minh	Vietnamese	Nguyễn Văn Minh	グエン・ヴァン・ミン
Park-Ji Hoon	Korean	박지훈	パク・ジフン

Notes: Native scripts are shown where applicable; Katakana is used to standardise the rendering of non-Japanese names.

The Western-Japanese profile is included to distinguish perceived responses to foreignness from responses to partial cultural or ethnic proximity. Unlike the Western profile, the Western-Japanese applicant combines a Western given name with a Japanese surname, thereby signalling both foreign association and potential insider status. This allows us to examine whether perceived rental access follows a binary Japanese/non-Japanese distinction or more graduated ethnic boundaries. Further details on name construction, validation procedures, script standardisation, and translation protocols are provided in Appendix C.

Because every profile is an equally fictitious, matched construction, no respondent's assessment of any given profile can be benchmarked against a true or independently verifiable outcome for that applicant. Instead, what the design measures is the belief itself, a point we return to in subsubsection 4.5.2.

3.3 Questionnaire Design

The questionnaire comprised three sections. The first collected socio-economic characteristics. The second asked respondents to evaluate the likelihood that each applicant would secure a rental viewing for a standardised property. The third, administered only to foreign-born respondents, included a condensed version of the Everyday Discrimination Scale (Williams et al., 1997), adapted to capture experiences relevant to rental access and search strategies.

The survey was developed in English and translated into Japanese, followed by verification by four native speakers. Questions on socio-economic characteristics were drawn from established government sources.[22] The main covariates include income, employment status, housing status, gender, age, ethnicity, and education. For foreign-born respondents, we additionally collect visa status, length of residence, and Japanese-language proficiency.

Perceived Likelihood of Securing a Rental Viewing. Respondents were shown a rental listing together with the applicant profiles described in subsection 3.2. For each profile, respondents stated the perceived likelihood that the applicant would be invited to a viewing. Respondents were also asked to assess the probability that they themselves would receive a viewing invitation. The same property listing was used throughout the experiment.

Perceived likelihoods were elicited on an 11-step scale, as illustrated in Figure 2. The property listing was sourced from *LIFULL Homes*,[23] one of Japan's largest real-estate platforms and the platform used by Sugasawa and Harano (2023). The selected property is a one-room apartment located in the Tokyo ward with the median number of foreign-born residents. The listing was chosen to match the profile of a single male applicant and to anchor respondents' expectations in a realistic rental-search environment (Weiss and Roberts, 2018).

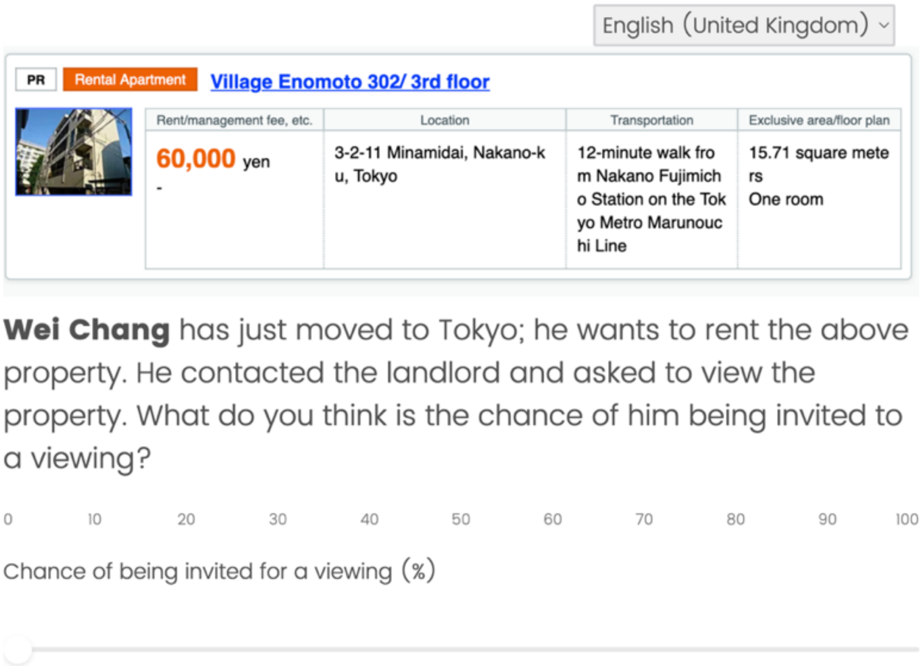
By eliciting a behavioural expectation about a named third-party applicant, rather than an attitudinal self-report about the respondent's own views, this format is comparatively unobtrusive relative to conventional prejudice measures, which are vulnerable to social-desirability pressure once respondents recognise a question is testing them for prejudice. This allows us to elicit native Japanese respondents' beliefs about non-Japanese applicants' prospects, i.e., hetero-stereotype, directly, a comparison rarely attempted in the existing discrimination-perception literature subsection 4.3.

Everyday Discrimination Scale. Foreign-born respondents were additionally asked four questions from the Everyday Discrimination Scale, reported in Appendix D. The items capture experiences of discourtesy, poorer service, being feared, and being viewed as dishonest. These

[22] The questions were taken from the *Ministry of Justice Basic Foreign Residents Survey* (`https://www.moj.go.jp/isa/content/001421198.pdf`), the *2020 National Census* (`https://www.stat.go.jp/english/data/kokusei/2020/summary.html`), and the *National Survey of Family Income, Consumption and Wealth* (`https://www.stat.go.jp/english/data/zenkokukakei/index.html`).

[23] See `https://www.homes.co.jp/`.

Figure 2: Eliciting the Perceived Likelihood to be Invited



Notes: The figure shows a snapshot of the experimental screens presented to participants to elicit their subjective perceptions of the likelihood of being invited to a viewing.

dimensions are relevant to landlord and agent screening decisions in rental markets (Clark, 2007; Hamovitch et al., 2022; Hanson et al., 2011). Responses are coded on a linear scale from 5 (almost every day) to 0 (never) and averaged into a composite index of experienced everyday discrimination.

We use this measure to examine whether prior experiences of discrimination are associated with respondents' perceived likelihood of securing a viewing invitation. We also relate the EDS score to integration-related variables, including permanent residency status, length of stay, and Japanese-language proficiency.

4 Empirical Analyses

4.1 Empirical Roadmap

In this section, we assess both the experimentally collected data[24] and complementary contextual evidence used to assess the public visibility of nationality-related screening signals.

Table 2 organises the analysis by evaluator and object of evaluation. We first establish the baseline perceived penalties reported by foreign-born and native Japanese respondents and then document whether nationality-related screening signals remain publicly visible. The next two subsections estimate PGDD and GGDD, respectively. A final subsection reports heterogeneity and potential informational mechanisms, including associations with prior discrimination, socio-economic characteristics, and Japanese-language proficiency.

The empirical sections distinguish descriptive results from interpretation. They establish the

[24]Table 12 in the Appendix reports summary statistics.

Table 2: Evaluation Strategy

	Foreign-born respondents	Section	Native Japanese respondents	Section
Prospects of non-Japanese applicants	Auto-stereotype	4.2	Hetero-stereotype	4.3
Own prospects	Personal evaluation	4.5.1	Native self-evaluation	4.5.1
Derived Discrepancies				
PGDD	Personal evaluation vs. auto-stereotype	4.5.1	–	–
GGDD	Auto-stereotype vs. hetero-stereotype	4.5.2	Auto-stereotype vs. hetero-stereotype	4.5.2
Benchmark self-evaluation gap	Personal evaluation vs. matched applicant profile	4.5.1	Native self-evaluation vs. Japanese applicant	4.5.1

Notes: PGDD denotes the Personal-Group Discrimination Discrepancy and compares foreign-born respondents' evaluations of their own prospects with their evaluations of the prospects of non-Japanese applicants. GGDD denotes the Group-Group Discrimination Discrepancy and compares foreign-born and native Japanese respondents' evaluations of the prospects of the same non-Japanese applicants. Native Japanese self-evaluations are used as a benchmark for general self-optimism, comparing native Japanese respondents' own expected viewing probability with their evaluation of the Japanese applicant.

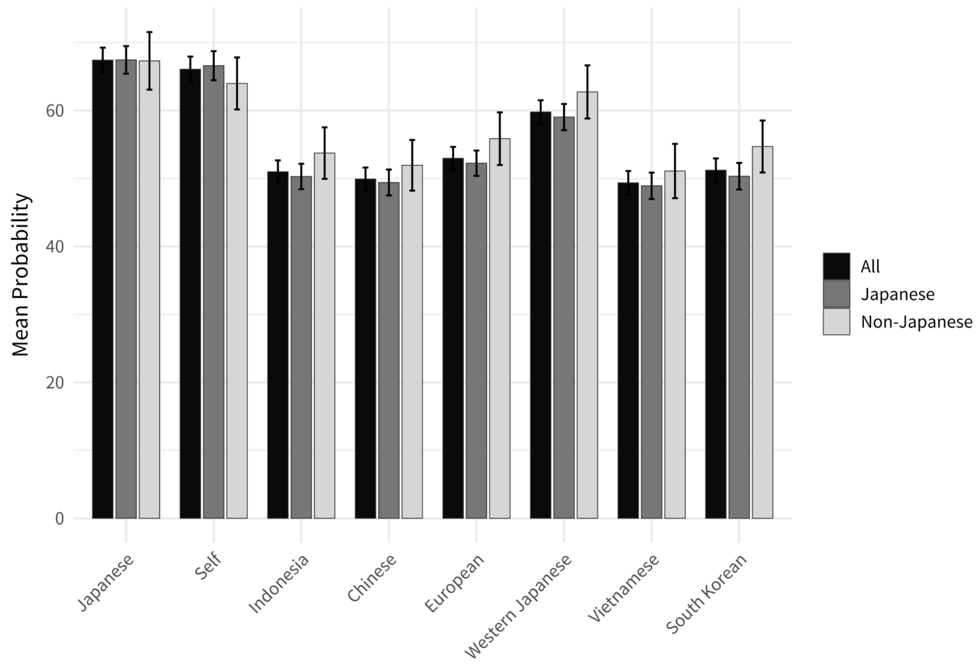
direction and magnitude of perceived access gaps and report correlational heterogeneity, but they do not identify which respondent group's beliefs are more accurate or establish causal mechanisms. Those questions are considered separately in section 5.

4.2 Auto-stereotypes of Foreign Disadvantage

We begin by examining auto-stereotypes, that is, foreign-born respondents' perceptions of the treatment of non-Japanese applicants. Across all applicant profiles, foreign-born respondents assign lower viewing probabilities to non-Japanese applicants than to the otherwise identical Japanese applicant (see Figure 3). The perceived penalties range from 4.6 percentage points for the Western-Japanese applicant to 16.2 percentage points for the Vietnamese applicant, with all differences statistically significant at the 1 per cent level.

To assess the substantive magnitude of these perceived disadvantages, we compute Cohen's d relative to the Japanese benchmark. The resulting effect sizes are moderate for most non-Japanese applicants. Chinese and Indonesian applicants are evaluated 0.59 and 0.55 standard deviations below the Japanese benchmark, respectively, while the Vietnamese applicant displays a similarly large effect size ($d = 0.57$). By contrast, the Western-Japanese applicant yields a substantially smaller effect size ($d = 0.21$). These estimates indicate that foreign-born respondents themselves perceive meaningful disadvantages facing non-Japanese applicants in Tokyo's rental market. At the same time, the comparatively small penalty for the Western-Japanese applicant suggests that perceived disadvantage is not organised solely around a binary Japanese/non-Japanese distinction. Instead, foreign-born respondents appear to perceive gradations in rental-market access, with profiles signalling partial cultural proximity to the Japanese majority evaluated more favourably than other non-Japanese profiles.

Figure 3: Mean stated viewing probabilities.



Notes: Collectively stated average probabilities of applicants of different ethnic backgrounds being offered a viewing. Overall probabilities as well as splits by native Japanese and foreign-born participants. Bars indicate standard deviations.

4.3 Hetero-stereotypes of Foreign Disadvantage

Assessing the remaining profiles, Figure 3 reveals that Japanese respondents expect *all* non-Japanese applicants to face a significantly lower probability of securing a viewing invitation than the Japanese applicant, with all differences statistically significant at the 1 per cent level. Among Japanese respondents, effect sizes are moderate to large ($d > 0.5$), with Indonesian and Chinese applicants positioned 0.71 and 0.67 standard deviations below the Japanese benchmark, respectively.

The largest perceived disadvantages are observed for Vietnamese, Chinese, Indonesian, and South Korean applicants, while the Western applicant is evaluated somewhat more favourably. The Western-Japanese applicant receives the most favourable evaluation among all non-Japanese profiles.

Eliciting hetero-stereotype directly is comparatively rare in the discrimination-perception literature, in part because direct measures of majority prejudice are vulnerable to social-desirability pressure, whereby respondents adjust their answers toward what is normatively acceptable once they recognise a question is testing them for prejudice. [25] Our survey design instead elicits a behavioural expectation about a named fictitious applicant, namely, the likelihood that this person receives a viewing invitation, rather than an attitudinal self-report about whether the respondent would invite them. Because respondents are not asked to characterise non-Japanese people

[25] This difficulty may have been less acute for the earliest studies of majority beliefs about out-groups, such as Katz and Braly (1933), classic elicitation of ethnic trait stereotypes attached to names among Princeton students.

directly, but merely to predict a market outcome for an individual applicant, our instrument is designed to be unobtrusive and therefore less susceptible to social-desirability adjustment than conventional prejudice measures. This enables us to compare robustly auto-stereotype and hetero-stereotype, a comparison the following sections take up in turn.

Comparing the auto-stereotype and hetero-stereotype results directly, both foreign-born and native Japanese respondents expect every non-Japanese profile to face lower viewing probabilities than the Japanese applicant. This directional agreement shows that perceived non-Japanese disadvantage is not confined to the affected group. It is shared across respondent groups, even though they differ in the magnitude of the penalty they assign. Such shared expectations may matter for behaviour on both sides of the market: applicants may avoid listings or providers they expect to be unreceptive, while landlords or agents may scrutinise applicants whom they expect others to regard as less acceptable.

4.4 Public Visibility of Nationality-Based Screening

The preceding analysis establishes that both foreign-born and native Japanese respondents expect non-Japanese applicants to face lower access probabilities. To connect these perceptions to the institutional environment, we examine whether nationality-related screening signals are publicly observable. This exercise supplies contextual and external-validity evidence; it does not measure realised treatment or estimate the prevalence of discriminatory wording across the full population of rental listings. We draw on two complementary sources.

First, qualitative evidence from journalistic reporting documents examples of discriminatory treatment toward foreign residents in the housing sector. Shaun McKenna's investigation for *The Japan Times* documents repeated cases in which foreign residents were denied housing applications solely on the basis of nationality, despite meeting standard financial and contractual requirements.[26] It further reports that real estate agents frequently justify these exclusions by citing landlords' concerns about communication barriers, cultural differences, or perceived risks associated with non-Japanese tenants. Importantly, the article emphasizes that such practices persist partly because Japan lacks comprehensive anti-discrimination legislation specifically prohibiting nationality-based exclusion in the housing market. Interviews with affected residents and housing advocates describe nationality-based screening as a recognised obstacle for foreign residents seeking accommodation in major urban areas.

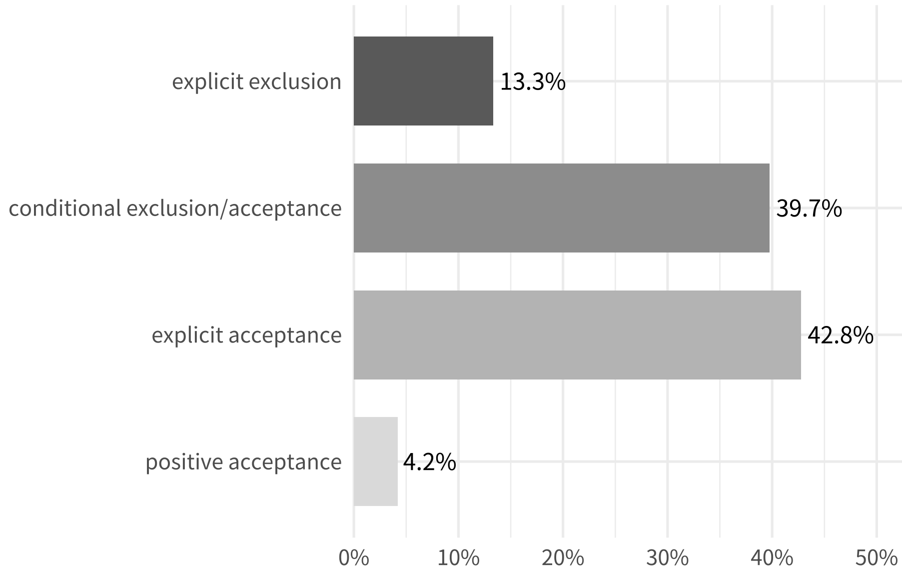
Second, we provide quantitative evidence by a visibility-based method examining how references to foreign tenants appear in publicly discoverable search across major Japanese real estate platforms: SUUMO, LIFULL HOME'S, and at home.[27] During our observation window between May 22–24, 2026, the three platforms jointly hosted more than 14.8 million listings, with SUUMO being the largest platform (approximately 7.56 million listings), followed by HOME'S (5.02 million) and at home (2.24 million). Using Google site-restricted searches conducted from a UK IP address, we searched the free-text content of listings for multiple Japanese-language terms associated with the explicit exclusion, consideration or explicit inclusion of foreign applicants, including **外国籍不可** ("foreign nationals not allowed"), **外国人不可** ("foreigners not allowed"), **外国籍相談** ("foreign nationals negotiable"), **外国人相談** ("foreigners negotiable"), **外国籍可** ("foreign nationals allowed"), **外国人可** ("foreigners allowed"), **外国籍歓迎** ("foreign nationals welcome"), and **外国人歓迎** ("foreigners welcome"), both nationwide and for the

[26] Shaun McKenna, "What's Behind Housing Discrimination in Japan?," *The Japan Times*, May 31, 2021, `https://www.japantimes.co.jp/community/2021/05/31/how-tos/housing-discrimination-still-problem-japan/`.

[27] SUUMO: `https://suumo.jp`; HOME'S: `https://www.homes.co.jp`; at home: `https://www.athome.co.jp`.

Greater Tokyo area.

Figure 4: Distribution of Nationality-Related Rental Listing Search Results



Notes: The Figure aggregates search terms cross platforms and into four categories: explicit exclusion, conditional exclusion/acceptance, explicit acceptance, and positive acceptance. Shares are calculated relative to all relevant search results returned by the platform-specific Google searches.

Among those adverts that do contain such references, we can therefore assess the relative prevalence of exclusionary, conditional, and explicitly inclusive language. Results shown in Figure 4 describe the composition of nationality-related search results returned by the visibility exercise. Of these results, 13.3% contained clearly exclusionary wording, 39.7% expressed conditional acceptance, and 46.9% were unconditionally open to foreign tenants. These shares should not be interpreted as estimates for all rental listings. Their relevance is instead that explicit and conditional nationality-related screening remains publicly discoverable on major housing platforms.

4.5 Disagreement about Discrimination

The relationships between the perceptual discrepancies identified in this study may be summarised as follows:

$$\text{Hetero-stereotype} \overset{\text{GGDD}}{>} \text{Auto-stereotype} \overset{\text{PGDD}}{>} \text{Personal disadvantage.}$$

The discrepancy between auto-stereotypes and perceived personal disadvantage corresponds to the classic Personal-Group Discrimination Discrepancy (PGDD), while the discrepancy between hetero-stereotypes and auto-stereotypes is what we term the GGDD. The existence of both, PGDD and GGDD, is particularly difficult to explain through personal experience alone.

4.5.1 Personal-Group Discrimination Discrepancies (PGDD)

The preceding analyses show that respondents evaluate their own prospects differently from the prospects of the matched applicant profile, and that this relationship is robust to socio-economic controls. To assess whether this pattern reflects a minority-specific personal-group discrepancy or a more general tendency toward favourable self-assessment, we use native Japanese respondents as a benchmark.

Specifically, for native Japanese respondents, we define the benchmark gap as the respondent's self-assessed viewing likelihood minus the stated viewing likelihood of the Japanese applicant. For foreign-born respondents, we define the corresponding matched self-evaluation gap as the respondent's self-assessed viewing likelihood minus the stated viewing likelihood of the applicant profile corresponding to the respondent's nationality, where available.

Table 4 reports the results. Among native Japanese respondents, the mean self-evaluation gap is statistically indistinguishable from zero (p-value = 0.230). Thus, native Japanese respondents do not rate their own prospects more favourably than those of the Japanese applicant. By contrast, among foreign-born respondents with a matched applicant profile, the mean self-evaluation gap is 9.10 percentage points and highly statistically significant (p-value < 0.001). The difference between the foreign-born and native Japanese gaps is hence positive and strongly significant (p-value < 0.001).

These results hence suggest that the personal-group discrepancy documented above is not simply driven by a general tendency of respondents to evaluate themselves more favourably than comparable applicants. Instead, the self-evaluation gap is concentrated among foreign-born respondents. The benchmark therefore strengthens the interpretation of the result as a minority-specific personal-group discrepancy.

Table 3: Group Perceptions vs. Individual Expectations

	(1)	(2)	(3)	(4)	(5)	(6)
	Log-log	Linear	Log-log	Linear	Log-log	Linear
Intercept	0.224	-6.537	1.236***	12.300***	-0.037	-15.042
	(0.401)	(12.062)	(0.102)	(2.059)	(0.400)	(12.027)
Matched nationality profile	0.647***	0.739***	0.643***	0.722***	0.643***	0.732***
	(0.024)	(0.024)	(0.023)	(0.024)	(0.024)	(0.024)
Vietnam	1.238***	22.346			1.321***	24.994*
	(0.461)	(13.981)			(0.456)	(13.810)
China	1.417***	32.978***			1.488***	34.893***
	(0.406)	(12.286)			(0.402)	(12.144)
Japan	1.164***	23.199*			1.259***	26.159**
	(0.400)	(12.132)			(0.396)	(11.992)
South Korea	1.233***	25.352**			1.299***	27.324**
	(0.419)	(12.677)			(0.414)	(12.522)
Log income			0.097***	2.801***	0.102***	2.921***
			(0.026)	(0.794)	(0.026)	(0.790)
Degree			0.082*	3.820**	0.063	3.085**
			(0.049)	(1.497)	(0.050)	(1.498)
Nationality FE	✓	✓			✓	✓
SE Controls			✓	✓	✓	✓
Adj. R^2	0.462	0.513	0.465	0.518	0.474	0.525
F-statistic	155.50***	190.20***	261.80***	322.50***	116.90***	143.10***
Observations	900	900	900	900	900	900

Notes: The dependent variable is respondents' self-perceived likelihood of receiving a viewing invitation. In log-log models, the dependent variable and matched nationality profile are transformed as $\log(1+x)$. The matched nationality profile is the perceived viewing likelihood for the applicant profile corresponding to the respondent's own nationality. The omitted nationality category is Indonesia. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The relationship between group-level and personal expectations remains stable after accounting for respondents' socio-economic characteristics. Returning to the regression specifications in Table 3, the coefficient on the matched nationality profile is nearly unchanged when individual

Table 4: Self-Evaluation Benchmark

	N	Mean gap	SE	t-statistic	p-value
Native Japanese	773	-0.95	0.79	-1.20	0.230
Foreign-born	127	9.10***	2.08	4.39	< 0.001
Difference: foreign-born – native Japanese	900	10.06***	2.13	4.72	< 0.001

Notes: The gap is defined as respondents' self-assessed viewing probability minus their evaluation of the matched applicant profile. For native Japanese respondents, the matched profile is the Japanese applicant. For foreign-born respondents, the matched profile is the applicant corresponding to the respondent's nationality, where available. The difference is estimated from a regression of the self-evaluation gap on an indicator for foreign-born respondents.

controls are added. This indicates that the observed gap between personal and group-level expectations is not driven by observable characteristics such as income or education. Among the control variables, higher income is positively and significantly associated with more optimistic self-assessed viewing probabilities in models (3)–(6). The effect of education is positive but weaker and less robust across specifications. Nationality fixed effects in models (1), (2), (5), and (6) reveal level differences across groups, but do not alter the central relationship of interest. Overall, these results suggest that the personal-group discrepancy is not simply an artefact of socio-economic composition across respondent groups.

4.5.2 Group-Group Discrimination Discrepancy (GGDD)

While both foreign-born (auto-stereotype) and native Japanese respondents (hetero-stereotype) perceive substantial disadvantages facing non-Japanese applicants, the magnitude of the perceived penalty is consistently larger among native Japanese respondents.

The estimates in Figure 3 indicate that Japanese respondents perceive stronger disadvantages for non-Japanese applicants than foreign-born respondents. Consistent with these univariate comparisons, the results indicate that native respondents assign even lower likelihood to non-Japanese applicants. Model (2) confirms again that Japanese respondents perceive a significantly larger penalty for non-Japanese applicants than non-Japanese respondents.

While conceptually related to the PGDD, this phenomenon concerns a different comparison. Both involve divergences in perceptions of disadvantage. However, whereas PGDD compares perceptions of personal disadvantage with perceptions of disadvantage facing one's own group, the present phenomenon compares perceptions of disadvantage facing the same minority group across majority and minority respondents. To our knowledge, this specific comparison, i.e., perceived market-access disadvantage facing an experimentally matched, fictitious target, assessed by both majority and minority raters, has not previously been examined, and we introduce a new term: *Group-Group Discrimination Discrepancy* (GGDD).

This comparison bears a structural resemblance to Judd and Park (1993), which likewise compares in-group and out-group raters' assessments of the same target group. Judd and Park (1993)'s central concern, however, is accuracy, i.e., whether a rater's perception of a group corresponds to some independently measurable criterion, typically the group's own self-reported attributes or behaviour.

In contrast, our survey design does not attempt an accuracy comparison of this kind. Applicant profiles in our experiment are fictitious constructions that hold every characteristic constant except an ethnically identifiable name. Accordingly, there is no independent criterion against which either hetero-stereotype or auto-stereotype could be scored as more or less accurate.

Both are beliefs about how a hypothetical, otherwise identical applicant would be treated, not estimates of some pre-existing traits or outcomes that could be independently verified.

This design allows us to compare instead the shared belief itself, without the accuracy problem attaching to its content. Market access is not a fixed criterion existing independently of belief, but is itself shaped by belief, which is therefore economically consequential. shapes market access and, hence, is economically consequential, Asking which of the two perceptions is closer to the truth would be the wrong question for our purposes. The relevant question is not whether native Japanese or foreign-born respondents perceive non-Japanese disadvantage more accurately, but how and why their beliefs about the very same hypothetical applicant diverge so systematically in magnitude, and what that divergence itself implies for how market friction is sustained and recognised.

The relationship between group-level and personal expectations remains stable after accounting for respondents' socio-economic characteristics. Returning to the regression specifications in Table 3, the coefficient on the matched nationality profile is nearly unchanged when individual controls are added. This indicates that the observed gap between personal and group-level expectations is not driven by observable characteristics such as income or education. Among the control variables, higher income is positively and significantly associated with more optimistic self-assessed viewing probabilities in models (3)–(6). The effect of education is positive but weaker and less robust across specifications. Nationality fixed effects in models (1), (2), (5), and (6) reveal level differences across groups, but do not alter the central relationship of interest. Overall, these results suggest that the personal-group discrepancy is not simply an artefact of socio-economic composition across respondent groups.

Beyond the overall magnitude of perceived disadvantage, GGDD may also be reflected in the way respondents map disadvantage across minority groups. The relative hierarchy assigned to different applicant profiles provides insight into how majority and minority respondents draw the social boundaries separating majority and minority groups. We therefore examine the relative hierarchy assigned to different applicant profiles.

Splitting responses by applicants' ethnic background reveals substantial heterogeneity, as shown in Figure 3. Japanese respondents expect all non-Japanese applicants to face a significantly lower probability of receiving a viewing invitation than the Japanese applicant, with all differences statistically significant at the 1% level. The largest perceived disadvantages are observed for Vietnamese, Chinese, Indonesian, and South Korean applicants, while the Western applicant is evaluated somewhat more favourably. The Western-Japanese applicant receives the most favourable evaluation among all non-Japanese profiles. This pattern suggests that perceived housing access is shaped not only by foreignness itself but also by signals of cultural proximity and partial insider status.

Assessing the relative ranking of applicant groups, native Japanese and foreign-born respondents exhibit broad agreement regarding the most favourably evaluated categories. Differences between the two groups emerge only among the lowest-ranked categories. Japanese respondents rank Chinese applicants above Vietnamese and Indonesian applicants, assigning Indonesians the lowest position overall. In contrast, foreign-born respondents rank Indonesians above Vietnamese and Chinese applicants, while Chinese applicants receive the lowest ranking in this group.

Nationality-specific analyses reported in Table 13 in the Appendix reveal substantial heterogeneity within the foreign-born sample. While respondents differ considerably in their relative evaluations of Chinese, Vietnamese, Indonesian, and South Korean applicants, the comparatively favourable assessment of the Western-Japanese applicant is remarkably consistent across respondent nationalities. Overall, these findings suggest that perceived housing access is structured less by a universally shared hierarchy among foreign groups than by broad distinctions in perceived cultural proximity and insider status. The stable advantage attributed to the Western-Japanese profile therefore suggests that respondents perceive access through graded boundaries of belonging rather than a simple Japanese/non-Japanese divide.[28]

Table 5: Differences in Perceived Likelihood of Viewing Invitations by Origin

	Indonesian	Chinese	Western Nationals	Western Japanese	Vietnamese	South Korean	Self-Perceived
Japanese							
Probability (%)	50.28	49.40	52.24	59.03	48.91	50.33	66.58
Difference from Japanese (%)	-17.12***	-17.90***	-15.16***	-8.44***	-18.59***	-17.18***	-0.95
Cohen's d	0.71	0.67	0.62	0.40	0.69	0.65	0.04
N	788	785	783	780	776	772	773
Non-Japanese							
Probability (%)	53.73	51.93	55.85	62.72	51.09	54.69	63.97
Difference from Japanese (%)	-13.57***	-15.57***	-11.48***	-4.61***	-16.24***	-12.64***	-3.36
Cohen's d	0.55	0.59	0.45	0.21	0.57	0.47	0.11
N	198	197	195	195	195	195	195

Notes: Entries report paired comparisons between the Japanese applicant (Taro Yamamoto) and each target applicant. "Probability" gives the mean perceived likelihood of receiving a viewing invitation for the target applicant. "Difference from Japanese" is calculated as target applicant probability minus Japanese applicant probability; negative values therefore indicate a lower perceived likelihood than for the Japanese applicant. Cohen's d is based on the paired difference between the Japanese applicant and the target applicant. N varies slightly across comparisons because each estimate uses respondents with complete paired responses for the relevant applicant pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

As shown in Figure 5, all non-Japanese profiles are evaluated less favourably than the Japanese benchmark. The most striking feature of the ranking is the stable position of the Western-Japanese applicant immediately below the Japanese benchmark in both respondent groups, making it the highest-ranked non-Japanese profile. Both groups further evaluate the Western applicant and the South Korean applicant relatively favourably compared with the remaining non-Japanese categories.

The Western-Japanese profile combines a Western given name with a Japanese surname, signalling both foreign association and partial insider status. The comparatively favourable evaluation of the Western-Japanese applicant is theoretically important because it suggests that perceived rental access is not governed solely by a binary distinction between Japanese and non-Japanese applicants (see also Pager and Shepherd, 2008). Rather, respondents appear to distinguish between varying degrees of cultural proximity and social closeness to the Japanese majority. This pattern is consistent with research showing that discrimination often operates through socially graded boundaries of perceived cultural proximity and assimilability rather than through simple majority-minority distinctions.

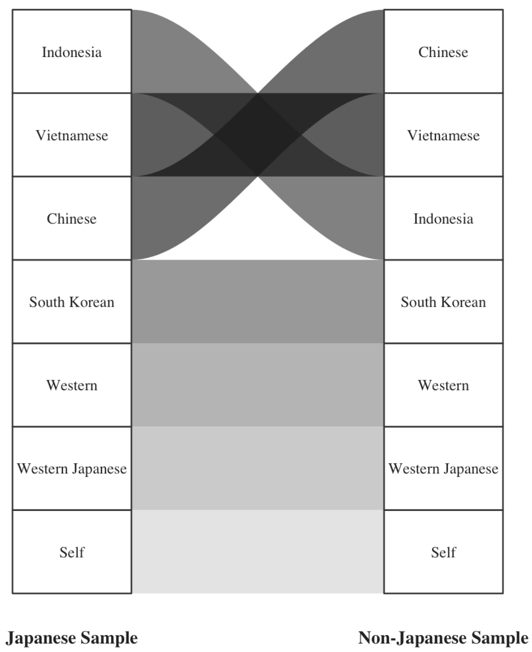
Differences between the two groups emerge only among the lowest-ranked categories. Japanese respondents rank Chinese applicants above Vietnamese and Indonesian applicants, assigning

[28] As an additional robustness check, we reweighted the full sample so that the nationality composition of survey respondents matches the corresponding population shares of Tokyo residents by nationality, using data from the Tokyo Statistical Yearbook (2024), `https://www.toukei.metro.tokyo.lg.jp/tnenkan/2024/tn24q3e002.htm`. The resulting estimates are qualitatively identical to those obtained for the native population. This similarity is unsurprising given the relatively small share of foreign residents in Tokyo. Consequently, the results based on the native population provide a close approximation to the perceptions of Tokyo's population as a whole.

Indonesians the lowest position overall. In contrast, foreign-born respondents rank Indonesians above Vietnamese and Chinese applicants, while Chinese applicants receive the lowest ranking in this group.

Nationality-specific analyses reported in Table 13 in the Appendix reveal substantial heterogeneity within the foreign-born sample. While respondents differ considerably in their relative evaluations of Chinese, Vietnamese, Indonesian, and South Korean applicants, the comparatively favourable assessment of the Western-Japanese applicant is remarkably consistent across respondent nationalities. Overall, these findings suggest that perceived housing access is structured less by a universally shared hierarchy among foreign groups than by broad distinctions in perceived cultural proximity and insider status. The stable advantage attributed to the Western-Japanese profile therefore suggests that respondents perceive access through graded boundaries of belonging rather than a simple Japanese/non-Japanese divide.

Figure 5: Ranking Shifts in Cohen's d by Origin



Notes: Rankings are based on Cohen's d from paired comparisons between the Japanese applicant and each comparison profile in perceived likelihood of viewing an invitation. Rank 1 (at the bottom) indicates the smallest negative deviation. The left column shows native Japanese respondents; the right column shows non-Japanese respondents.

In addition to the univariate comparisons, we test for systematic differences in a multivariate regression framework (Table 6). Model (1) regresses stated viewing probabilities on applicant-profile indicators, with the Japanese applicant serving as the reference category. The estimated coefficients closely mirror the averages reported in Table 5. Since each respondent evaluates multiple applicant profiles, standard errors are clustered at the respondent level throughout.

Model (2) adds respondent fixed effects. This specification absorbs respondent-specific baseline differences in stated probabilities, such as a general tendency to assign higher or lower viewing chances across all profiles. The coefficients are therefore identified from within-respondent differences in perceived access across applicant profiles. The inclusion of respondent fixed effects leaves the estimated applicant-profile penalties virtually unchanged, reinforcing the robustness of the main finding that all non-Japanese profiles are perceived to face lower access probabilities than the Japanese benchmark.

Table 6: Perceived Likelihood of Viewing Invitations

	(1)	(2)	(3)
Constant	67.406***		
	(0.935)		
Chinese	-17.500***	-17.387***	-15.458***
	(0.851)	(0.849)	(1.891)
Western Nationals	-14.446***	-14.349***	-11.401***
	(0.799)	(0.793)	(1.830)
Indonesia	-16.430***	-16.407***	-13.566***
	(0.776)	(0.775)	(1.763)
South Korean	-16.199***	-16.117***	-12.555***
	(0.857)	(0.848)	(1.929)
Vietnamese	-18.055***	-17.999***	-16.160***
	(0.878)	(0.870)	(2.041)
Western-Japanese	-7.639***	-7.580***	-4.524***
	(0.684)	(0.675)	(1.540)
Self	-1.348*	-1.308*	-3.278
	(0.779)	(0.771)	(2.171)
Chinese × Japanese respondent			-2.414
			(2.116)
Western × Japanese respondent			-3.686*
			(2.030)
Indonesia × Japanese respondent			-3.555*
			(1.962)
South Korean × Japanese respondent			-4.458**
			(2.147)
Vietnamese × Japanese respondent			-2.300
			(2.256)
Western-Japanese × Japanese respondent			-3.821**
			(1.713)
Self × Japanese respondent			2.470
			(2.312)
Respondent FE	.	✓	✓
Clustered SE by respondent	✓	✓	✓
Adj. R^2	0.058	0.079	0.082
F-statistic	69.16***	238.85***	121.70***
Observations	7820	7820	7820

Notes: The dependent variable is the stated perceived likelihood of receiving a viewing invitation. The reference applicant is the Japanese applicant, Taro Yamamoto. Columns (2) and (3) include respondent fixed effects, absorbing respondent-specific baseline levels in stated probabilities. Standard errors, reported in parentheses, are clustered at the respondent level in all specifications. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Model (3) interacts the applicant-profile indicators with an indicator for native Japanese respondents. This specification tests whether native Japanese and foreign-born respondents differ in the perceived penalties they assign to the same applicant profiles. A Wald test rejects the joint null that all interaction terms are zero, indicating that perceived applicant-profile penalties differ systematically across respondent groups ($\chi^2_7 = 27.006$, $p = 0.0003$).

Consistent with the univariate comparisons in Table 5, native Japanese respondents generally assign larger penalties to non-Japanese applicants than foreign-born respondents. The additional penalty is most clearly visible for South Korean and Western-Japanese applicants, where the interaction terms are statistically significant at conventional levels. The interactions for Western and Indonesian applicants are negative and weakly significant, while the differences for Chinese and Vietnamese applicants are not statistically significant. Thus, the evidence suggests that majority and minority respondents broadly agree that non-Japanese applicants face disadvantages, but differ in the magnitude of these perceived disadvantages for selected profiles.

The Western-Japanese applicant is particularly informative. Foreign-born respondents evaluate this profile relatively close to the Japanese benchmark, whereas native Japanese respondents assign a significantly larger penalty. This suggests that native respondents may perceive partial foreignness more sharply, even when the applicant also signals cultural proximity through a Japanese surname. South Korean applicants also occupy an intermediate position in the perceived hierarchy: they are evaluated less favourably than Western-Japanese and Western applicants, but more favourably than Chinese, Vietnamese, and Indonesian applicants. This pattern may reflect the coexistence of cultural familiarity and historically strained relations between Japan and South Korea.

Overall, the regression results indicate a substantial degree of agreement between native Japanese and foreign-born respondents regarding the relative ordering of applicant profiles. Both groups place the Western-Japanese applicant closest to the Japanese benchmark and assign the largest penalties to other non-Japanese profiles. The principal difference between the two respondent groups therefore lies less in a fundamentally different perceived hierarchy than in the magnitude of the disadvantage attributed to particular applicant profiles. This pattern is consistent with the Group-Group Discrimination Discrepancy: majority and minority respondents recognise the same broad structure of disadvantage, but differ in how strongly they perceive it.

4.6 Potential Informational Mechanisms Giving Rise to Disagreement

The preceding subsections establish the two principal discrepancies. We now examine correlates that may help organise their interpretation. These analyses are descriptive and should not be read as causal tests of the mechanisms discussed later.

4.6.1 The Role of Foreign-Born Respondents' Own Experience of Discrimination

Additional questions elicit foreign-born respondents' self-reported experiences of discrimination ($N = 195$ valid), as detailed in subsection 3.3. Respondents are presented with four scenarios capturing how others behave towards them in different contexts. Responses are coded such that lower values correspond to more frequent experiences of unequal treatment. In addition to examining each dimension separately, we construct an individual-level index of experienced discriminatory behaviour by averaging responses across the four scenarios.

As shown in Figure 8 in the Appendix, only a small fraction of foreign-born respondents report never having experienced discriminatory behaviour. The modal response across all scenarios is "A few times a month" (see Table 7), indicating that recurring exposure to subtle forms of discrimination is common within the sample. These items capture everyday instances of unequal treatment adapted to the housing context and are expected to be closely related to respondents' perceived likelihood of receiving a viewing invitation.

Table 7: Most Frequent Response for Each EDS Item

Question	Most frequent response	n	%
EDS1	A few times a month	68	34.2
EDS2	A few times a month	85	42.7
EDS3	A few times a month	107	53.8
EDS4	A few times a month	119	59.8

Notes: Entries report the modal response category for each EDS item. Percentages are the share of responses for that item selecting the modal category.

To examine this relationship, we first estimate bivariate correlations between the stated probability of securing a viewing and the composite discrimination index. We find a positive yet modest correlation (Spearman's $\rho = 0.125$, $H_0: \rho = 0$ vs. $H_1: \rho > 0$ rejects H_0 with p-value = 0.041), indicating that stated perceived viewing likelihoods capture a related—though not identical—construct to broader measures of perceived discrimination. Repeating this test separately for each discrimination dimension reveals that no individual component is, in isolation, statistically significantly correlated with the subjectively stated viewing probability. This suggests that the observed association reflects the combined effect of multiple dimensions of perceived discrimination rather than any single category alone.

Next, we regress the stated probability of securing a viewing on the composite discrimination index and incrementally add further controls. We also test for a non-linear relationship between the two factors. Table 8 shows that respondents' perceived likelihood of receiving a viewing invitation is systematically associated with their prior experiences of discrimination. Across all six specifications (1)–(6), the EDS score enters positively and is highly statistically significant, while its squared term is negative and statistically significant throughout, confirming a robust nonlinear relationship. This implies that respondents with more frequent prior discrimination experiences (i.e., lower EDS scores) report lower expected probabilities of receiving a viewing invitation, whereas those who report less frequent discrimination express more optimistic expectations. The negative coefficient on the squared term further suggests diminishing marginal effects: the positive association between more favourable past treatment and perceived viewing chances weakens at higher levels of the EDS index.

Furthermore, we examine whether integration into Japanese society is associated with individuals' perceived success in the housing application process. Integration is proxied by permanent residency status, length of stay in Japan, and self-reported Japanese language proficiency. Length of stay, included in specifications (4) and (6), is statistically insignificant in both models, suggesting that time spent in Japan does not meaningfully explain variation in perceived probabilities of receiving a viewing invitation. Japanese language proficiency, introduced in specifications (2) and (6), is positively associated with perceived success and weakly significant in specification (2); however, the effect becomes statistically insignificant in specification (6) once additional controls are included, indicating limited robustness. Permanent residency status, included in specifications (3) and (6), also remains positive but statistically insignificant throughout.

In contrast, economic resources appear to matter. Log income, included in specifications (5) and (6), enters positively and is statistically significant in both models, suggesting that higher-

Table 8: Determinants of Respondents' Perceived Own Likelihood of Receiving a Viewing Invitation

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	41.045***	27.784**	38.136***	37.077***	32.674***	23.362*
	(8.864)	(11.265)	(9.054)	(12.427)	(9.463)	(13.524)
EDS score	20.550***	18.542***	19.667***	20.950***	21.081***	18.440***
	(6.735)	(6.775)	(6.741)	(6.806)	(6.662)	(6.853)
EDS score^2	-3.999***	-3.439***	-3.781***	-4.078***	-4.092***	-3.406**
	(1.281)	(1.307)	(1.286)	(1.295)	(1.267)	(1.328)
Language skills		1.441*				1.166
		(0.764)				(0.783)
Permanent visa			5.832			4.503
			(3.953)			(4.035)
Length of stay				0.565		-0.396
				(1.238)		(1.272)
Log income					4.897**	4.382**
					(2.090)	(2.111)
Adj. R^2	0.039	0.051	0.045	0.035	0.061	0.065
F-statistic	4.91***	4.50***	4.02***	3.33**	5.18***	3.25***
Observations	195	195	195	195	195	195

Notes: The dependent variable is respondents' perceived own likelihood of receiving a viewing invitation. *EDS score* measures the frequency with which respondents report having experienced discriminatory behaviour. *Language skills* is defined as the sum of reading and speaking ability. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

income respondents report more optimistic expectations regarding their rental prospects.

Overall, while the models explain only a small fraction of the variation in perceived viewing probabilities (adjusted R^2 ranging from 0.035 to 0.065 across specifications (1)–(6)), they consistently indicate that prior experiences of discrimination are a key determinant of respondents' expectations. Income also plays a meaningful role, whereas conventional indicators of integration—such as language proficiency, visa status, and length of stay—show limited and non-robust associations. By contrast, perceptions of group-level success, namely the collective perceived success of foreign-born respondents in the Japanese housing market, appear to be more strongly associated with perceived individual outcomes. Taken together, these findings suggest that subjective expectations in the rental housing market are shaped less by individuals' own integration characteristics and more by broader perceptions of how foreign-born respondents are treated within Japanese society, with economic resources playing a secondary but still important role.

4.6.2 The Role of Respondents' Socio-Economic Characteristics

Having established the existence and structure of GGDD, we next investigate whether respondents' characteristics, namely, their socio-economic statuses, are systematically associated with their evaluations of disadvantage facing non-Japanese applicants.

To focus on the overall perception of minority disadvantage, we next collapse all non-Japanese profiles into a single 'Non-Japanese Applicants' (NJA) category and examine heterogeneity by

Table 9: Heterogeneity Results

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	67.406***	67.293***	66.201***	64.805***	68.010***	64.165***
	(0.934)	(2.158)	(1.253)	(1.344)	(1.355)	(1.547)
NJA	-15.046***	-12.301***	-14.701***	-13.352***	-16.487***	-12.656***
	(0.739)	(1.658)	(0.995)	(1.042)	(1.037)	(1.234)
Japanese respondent		0.141				
		(2.393)				
NJA × Japanese respondent		-3.434*				
		(1.851)				
Female			2.911			
			(1.873)			
NJA × Female			-0.828			
			(1.478)			
Age > 50				5.263**		
				(1.860)		
NJA × (Age > 50)				-3.504**		
				(1.474)		
Income = Below Median					-1.189	
					(1.870)	
NJA × (Income = Below Median)					2.842*	
					(1.474)	
Degree						5.417**
						(1.932)
NJA × Degree						-4.005**
						(1.533)
Clustered SE by Respondents	✓	✓	✓	✓	✓	✓
Adj. R^2	0.035	0.036	0.036	0.037	0.035	0.036
F-statistic	248.80***	87.47***	86.58***	88.26***	84.85***	87.30***
Observations	6852	6852	6852	6845	6852	6852

Notes: The reference category for *NJA* is the Japanese applicant (Taro Yamamoto). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

respondents' characteristics. Table 9 reports the results.[29] Model (1) reaffirms the baseline findings obtained with separate non-Japanese applicant categories: Participants jointly expect a lower chance of NJAs to secure a viewing as compared to the Japanese applicant. Model (2) confirms again that Japanese respondents perceive a significantly larger penalty for non-Japanese applicants than non-Japanese respondents. Model (3) tests for differences by gender. Perceived discrimination appears not to differ between female and male participants.

Model (4) splits the sample at the median age. Older respondents expect higher viewing probabilities overall, but they also perceive a significantly larger disadvantage for NJAs. The findings for gender and age are in line with results from Luxembourg based on revealed discrimination rather than stated perceptions (Lepinteur et al., 2025). The finding that older Japanese participants perceive stronger disadvantages for NJAs may be driven by several mechanisms. The age gradient may reflect experience-based learning, cohort effects, or differential exposure to recent market developments. Distinguishing between these mechanisms is beyond the scope of the present design and remains an avenue for future research.

Model (5) indicates that lower-income respondents anticipate less discrimination against NJAs, implying that higher-income respondents perceive a stronger penalty. Model (6) distinguishes respondents by educational attainment (degree vs. no degree). Respondents with a degree perceive a larger disadvantage for non-Japanese applicants.

Compared with results for Luxembourg by Lepinteur et al. (2025), our findings show both clear parallels and some differences across the heterogeneity dimensions examined. Most importantly, Models (3) and (4) align with their results in showing no meaningful heterogeneity by gender and stronger perceived discrimination among older respondents. However, Model (6) reveals a different education gradient: whereas Lepinteur et al. (2025) find stronger discrimination among less-educated participants, our results suggest that respondents with a degree perceive a larger disadvantage for non-Japanese applicants.

To further investigate the education effect in Model (6), we estimate the specification on subsamples defined by participants' nationality. When including participants from Asian countries with comparatively strong education systems (Japan, China, Taiwan, and South Korea), the estimated interaction effect remains qualitatively similar but becomes smaller in magnitude.

We further test whether restricting the sample to native Japanese or foreign-born respondents alters the effects in Models (2)–(6). The results remain quite stable across these restricted samples, and we do not find substantially different estimates when focusing on either group separately.

Overall, respondent characteristics are systematically associated with the perceived magnitude of disadvantage attributed to non-Japanese applicants. Older respondents, higher-income respondents, and respondents with a university degree consistently assign larger disadvantages to non-Japanese applicants, whereas the estimated effects are otherwise stable across respondent subgroups.

4.6.3 The Role of Foreign-Born Respondents' Linguistic Proficiency

We next examine whether GGDD varies systematically with foreign-born respondents' Japanese language proficiency. Specifically, we calculate the gap between the stated viewing likelihood of the Japanese applicant and the average stated viewing likelihood of non-Japanese applicants,

[29] We cluster standard errors at the respondent-level. Controlling for individual FE again leads to very similar marginal effects.

excluding the respondent's own profile type and the Western-Japanese profile.[30] Larger values indicate a larger perceived Japanese-applicant advantage.

The results suggest a threshold pattern rather than a smooth linear association (see Appendix E.1 for details). Respondents in the lowest Japanese-proficiency categories perceive substantially smaller Japanese-applicant advantages than other foreign-born respondents. For reading, the lowest-proficiency group reports an average gap of -7.52 percentage points, compared with 14.89 points among all other respondents. Similar low-proficiency contrasts appear for speaking and for the combined reading-speaking score. Once the lowest-proficiency group is excluded, the remaining correlation is small and statistically insignificant. These results provide suggestive, qualified support for the linguistic-asymmetry interpretation: Japanese-language access appears most relevant at the lower end of the proficiency distribution, rather than operating as a general linear relationship across all levels of proficiency.

As a complementary test, we use a fixed Japanese–Chinese comparison. This provides a cleaner comparison because the non-Japanese reference applicant is held constant, avoiding changes in the outcome that may arise from averaging across heterogeneous foreign applicant profiles. The comparison is also substantively important. Chinese-origin respondents constitute the largest group among foreign-born participants in our sample, accounting for 44.8 per cent of foreign-born respondents. Chinese nationals also represent 38.0 per cent of Tokyo's registered foreign population, making them the largest foreign nationality group in the city.[31] Moreover, Chinese nationals form the largest group of foreign new arrivals in the migration data used to construct the applicant profiles. The Japanese–Chinese comparison therefore captures both a central minority group in Tokyo's rental market and, for Chinese respondents, an own-group evaluation relative to the Japanese benchmark.

This comparison is particularly informative because Chinese residents may have access to more extensive co-ethnic community infrastructure than smaller foreign groups, including social networks, information channels, and greater familiarity with housing-search strategies. If such infrastructure substantially insulated respondents from perceived market disadvantage, one might expect the language channel to be weaker in this setting. Instead, Japanese proficiency remains positively associated with the perceived Japanese-applicant advantage over the Chinese applicant. Using the combined reading–speaking score, the correlation is $\rho = 0.207$ among all foreign-born respondents (p-value=0.004) and remains substantively similar within Chinese respondents themselves ($\rho = 0.201$, p-value=0.063). These results suggest that greater Japanese-language proficiency is associated with perceiving a larger Japanese-applicant advantage even when the comparison involves respondents' own nationality group and a group likely to benefit from comparatively developed co-ethnic support structures.

Taken together, these results provide suggestive support for the linguistic-asymmetry interpretation. In the aggregate comparison across non-Japanese applicants, the association is concentrated among respondents with the lowest Japanese proficiency. In the fixed Japanese–Chinese comparison, the association is positive and remains substantively similar when restricted to Chinese respondents. This pattern is consistent with the idea that Japanese-language access shapes perceptions of nationality-based disadvantage, while also indicating that the strength and form of the association depend on the comparison group considered.

[30] We exclude the Western-Japanese applicant as foreign participants – unlike native participants – appear not to strongly differentiate between native and Western-Japanese profiles (see Figure 3).

[31] Tokyo Metropolitan Government, *Tokyo Statistical Yearbook 2023*, Table 2-4, "Foreign Residents by District and Nationality," available at `https://www.toukei.metro.tokyo.lg.jp/tnenkan/2023/tn23q3e002.htm`.

5 From Perceived Disadvantage to Policy: Information, Adaptation, and Regulatory Responses

The empirical analysis establishes three descriptive patterns: both respondent groups perceive non-Japanese disadvantage; foreign-born respondents perceive less disadvantage for themselves than for their broader nationality group; and native Japanese respondents perceive a larger group-level penalty than foreign-born respondents do. These results identify PGDD and GGDD, but they do not determine which assessments are most accurate. The interpretations below therefore treat the heterogeneity results and the institutional evidence as suggestive mechanisms rather than causal tests, and draw out their implications for anti-discrimination regulation that relies primarily on individual complaints and completed discriminatory acts.

5.1 Interpreting PGDD: Epistemic Burden and Minority Adaptation

The personal-group comparison provides strong evidence of PGDD among foreign-born respondents. They recognise disadvantage facing applicants from their broader nationality group but assess their own prospects more favourably. The native Japanese benchmark is important: native respondents do not display a comparable premium when evaluating themselves relative to the Japanese applicant. The pattern is therefore unlikely to be explained solely by a general tendency toward favourable self-assessment.

One explanation is the greater epistemic burden involved in identifying discrimination against oneself. Rental outcomes are often opaque. A non-response, failure to receive a viewing invitation, or rejection may have several plausible explanations, making it difficult for an individual applicant to attribute a particular outcome confidently to nationality. Group-level disadvantage can be easier to recognise because it can be inferred from repeated incidents, information about others, and broader public signals rather than from a single ambiguous transaction.

A complementary explanation is behavioural adaptation. Foreign-born residents who expect barriers may acquire local knowledge, rely on informal networks, or direct their search toward landlords and agencies known to accept foreign tenants. They may therefore believe both that their group faces substantial barriers and that they personally know how to navigate them. The positive association between income and perceived personal access is consistent with this interpretation, because economic resources may increase an applicant's perceived capacity to overcome screening constraints. The association does not, however, establish that adaptation causes PGDD.

Under this interpretation, PGDD need not imply denial of group disadvantage. It may instead reflect a combination of attributional ambiguity and individually rational strategies for navigating a segmented market. These strategies can improve personal prospects while leaving the underlying restriction on the set of available dwellings unchanged.

5.2 Interpreting GGDD: Collective Information and Epistemic Asymmetry

GGDD concerns a different comparison. Native Japanese and foreign-born respondents agree on the direction of non-Japanese disadvantage, but native respondents assign a larger penalty to the same applicant profiles. Because the experiment does not observe landlords' realised decisions, the majority assessment cannot be treated as an accuracy benchmark. The result nevertheless shows that majority and minority respondents draw systematically different conclusions from the same hypothetical profiles.

Collectively available information offers one possible explanation. Public discussion, media coverage, and visible listing language may communicate that non-Japanese applicants occupy a disadvantaged position even to native Japanese respondents who do not personally face nationality-based exclusion. The platform exercise supports the narrower claim that such signals are publicly observable, not that they are representative of all listings.

Access to this information may be asymmetric. Much of Japan's housing-market information and public discussion is available primarily in Japanese. Among foreign-born respondents, those with the lowest Japanese-language proficiency perceive a substantially smaller Japanese-applicant advantage. This threshold pattern is consistent with an informational account in which limited access to Japanese-language signals attenuates group-level perceptions. Because language proficiency is not randomly assigned, the result remains correlational and may also capture selection, socio-economic resources, or unobserved differences in market experience.

The political and linguistic fragmentation of the foreign-born population may reinforce this informational asymmetry. Foreign residents comprise several nationality and language groups and generally lack direct electoral representation. These features may limit the formation of a single, widely articulated account of collective housing disadvantage. This interpretation is plausible in the Japanese institutional context, but it should be viewed as a hypothesis generated by the observed pattern rather than as an identified mechanism.

5.3 Policy Implications for Anti-Discrimination Regulation

These interpretive patterns point to four policy-relevant weaknesses in systems that rely primarily on individual complaints and completed discriminatory acts. PGDD may reduce recognition and reporting of personal discrimination; GGDD may weaken recognition of disadvantage as a collective problem; visible screening signals may discourage participation before a formal refusal occurs; and individual adaptation may segment the market rather than remove the underlying friction. Tokyo illustrates these mechanisms in a setting where nationality-based housing discrimination is addressed largely through general private-law claims and non-binding administrative measures.

5.3.1 PGDD and the Limits of Individual Enforcement

PGDD indicates that individuals may recognise disadvantage facing their group without identifying themselves as personally disadvantaged. In housing, opaque screening decisions intensify the attribution problem: applicants rarely observe why an enquiry was ignored or a viewing was refused. A complaint-based enforcement system therefore depends on precisely the recognition that PGDD may weaken, creating risks of under-reporting, under-claiming, and under-litigation.

This concern is particularly relevant in Japan, where nationality-based housing discrimination is generally pursued through case-specific private-law claims rather than a comprehensive statutory housing-discrimination regime. Even when proceedings are initiated, attributional uncertainty becomes an evidential burden. Clear statutory standards, accessible complaint procedures, and an independent body able to investigate patterns across cases would reduce reliance on individual applicants proving the motive behind a single opaque decision.

5.3.2 GGDD and the Limits of Collective Correction

GGDD suggests that affected groups may perceive the scale of collective disadvantage less strongly than majority respondents do. This can increase the organisational costs of systemic correction: individuals may recognise isolated barriers without viewing them as a sufficiently common group problem to justify cooperation, advocacy, or political mobilisation. The issue precedes the familiar collective-action problem because it concerns whether a shared interest is recognised in the first place.

Foreign residents' limited political incorporation can compound this informational constraint. Where the affected population has restricted electoral voice and is fragmented across languages and nationalities, systemic reform may generate limited direct political returns. Institutions that collect and publish comparable evidence on housing access can therefore matter not only for enforcement but also for establishing whether disadvantage is patterned rather than exceptional.

5.3.3 Regulating Screening Signals and Approval Criteria

The visibility exercise provides a rationale for addressing discriminatory advertisements and discouraging statements before disadvantage materialises in a completed transaction. Nationality-related wording can affect more than the landlord and applicant involved in a particular listing. Publicly observable signals may shape expectations about who is welcome to participate, leading applicants to self-select away from parts of the market and normalising differential screening among providers.

The exercise does not show that such wording is prevalent across all listings. It does show that explicit exclusion and conditional acceptance remain publicly discoverable on major platforms. Regulation could therefore require platforms and agents to remove explicitly exclusionary statements, standardise permissible screening criteria, and provide reporting channels for discriminatory content. Such rules would reduce both direct discouragement and the expressive signal that equal access cannot be assumed.

Evidence from later stages of the rental process reinforces the case for regulating screening criteria as well as publicly visible advertisements. Approval gaps persist even after accounting for predicted rent arrears and observable payment capacity, suggesting that nationality-based screening cannot be justified simply as conventional financial-risk assessment (Suzuki et al., 2022). Platforms, agents, landlords, and rent-guarantee providers should therefore be required to apply transparent and reviewable criteria, while enforceable restrictions on nationality-based exclusion could limit discretionary screening that is only weakly related to documented payment risk.

5.3.4 Adaptation, Segmentation, and Market Efficiency

Foreign residents may rationally concentrate their searches on agencies, landlords, and channels known to accommodate foreign tenants. Targeted searching can lower individual search costs and help explain why respondents view their personal prospects more favourably than their group's prospects. Yet successful navigation through specialised channels does not imply equal access to the market as a whole.

Adaptation may instead create persistent segmentation. Foreign and Japanese applicants can end up searching through partly separate networks, reducing the set of feasible matches and weakening competition across market segments. Differences in price, quality, and location may then persist, while less receptive providers face little pressure to change because foreign applicants learn to avoid them.

Policy should therefore assess access to the mainstream rental market, not merely whether some specialised channels serve foreign tenants. Greater transparency in screening criteria, monitoring of platform and agency practices, and enforceable equal-treatment standards would address the underlying restriction on choice rather than leaving applicants to develop private workarounds.

6 Conclusions

This paper studies perceived ethnic discrimination as a market-access friction in Tokyo's rental housing market. Using a name-based survey experiment, we elicit expected viewing probabilities for otherwise identical applicants and compare evaluations across native Japanese and foreign-born respondents. Both groups expect non-Japanese applicants to face lower access probabilities than the Japanese applicant, indicating that perceived disadvantage is not confined to those directly exposed to nationality-based exclusion.

Two discrepancies organise the results. First, foreign-born respondents assess their own prospects more favourably than those of applicants from their broader nationality group. The absence of a comparable self-evaluation premium among native Japanese respondents supports interpreting this result as a Personal-Group Discrimination Discrepancy. Second, native Japanese respondents assign larger penalties to the same non-Japanese applicants than foreign-born respondents do, producing a Group-Group Discrimination Discrepancy. The two patterns show that agreement about the direction of disadvantage can coexist with substantial disagreement about its magnitude.

The heterogeneity results provide suggestive evidence about how these beliefs may be formed. Prior experiences of everyday discrimination are associated with more pessimistic personal expectations, while income is associated with more favourable assessments of one's own prospects. Respondents with the lowest Japanese-language proficiency perceive a smaller Japanese-applicant advantage, consistent with an informational mechanism. These associations are not causal, and the study cannot determine which respondent group's beliefs most closely approximate realised landlord behaviour.

The paper contributes by shifting attention from realised differential treatment to beliefs about access, by providing evidence from a major non-Western rental market, and by distinguishing personal-group from majority-minority discrepancies in evaluations of the same market constraint. The platform exercise complements the survey by showing that nationality-related screening signals remain publicly visible, while not estimating their prevalence among all rental listings.

The results also clarify why case-by-case enforcement may be insufficient. Applicants may not identify ambiguous outcomes as discrimination, affected groups may not perceive disadvantage as a common problem, and visible screening signals may discourage applications before a refusal occurs. Individual adaptation can improve access for some residents but may also sustain market segmentation. Reducing these frictions therefore requires transparent screening standards and credible institutional signals that equal access applies across the mainstream rental market, rather than only within specialised channels.

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Appendix

A Historical Development of Tokyo's Rental Housing Market

This appendix provides additional historical and institutional background on the development of Tokyo's rental housing market.

A.1 Spatial Order under Tokugawa Japan and the Development of Rental Housing

The feudal order of Tokugawa Japan (1603–1868) had a lasting influence on Edo's urban structure and, by extension, on the development of what is now Tokyo's rental housing market. Edo was organised around the political authority of the Tokugawa shogunate and the spatial hierarchy of samurai and daimyo households. Daimyo were themselves differentiated into hierarchical categories, including *shimpan*, *fudai*, and *tozama*, with politically trusted groups located closer to the centre of power.

This spatial and political hierarchy reflected broader insider–outsider distinctions within Tokugawa society. These distinctions were embedded in household-based forms of social organisation and dependency, in which tenants, servants, and retainers could be incorporated into quasi-familial structures, while those outside such networks were positioned as social outsiders. Historical accounts suggest that this distinction continued to shape modern perceptions of non-Japanese residents as socially separate outsiders.

The concentration of daimyo and samurai households in Edo also contributed to the development of rental arrangements. Samurai land was not freely alienable in the same way as commercially transferable urban land in cities such as Osaka. As a result, financially constrained samurai households increasingly rented out land rather than transferring ownership. This practice later spread beyond samurai districts and contributed to the historically high prevalence of rental housing in Tokyo.

Following the Meiji Restoration and the Land Tax Reform, land became increasingly commodified and subject to taxation. These institutional changes further incentivised landlords to generate income through leasing while retaining ownership. At the same time, landownership continued to carry symbolic and status-related significance, particularly in former samurai districts. These historical developments contributed to the long-standing importance of rental housing in Tokyo.

A.2 Wartime Tenant Protection and Long-Term Tenure Security

The introduction of the *Building Protection Act*, the *Land Lease Act*, and the *Building Lease Act* before and after WWI was driven by rapid land price inflation, which incentivised landlords to transfer land to third-party to remove tenants whose leasehold interests lacked proprietary protection under Japanese civil law.

However, the decisive transformation of Japanese tenant protection occurred during the transition toward a total-war economy, following the Second Sino-Japanese War beginning in 1937. Against the background of wartime militarism, the government sought to stabilise urban labour supply for armament production and the war economy. Under the National Mobilisation Law (1938), the government introduced the *Rent Control Order* in October 1939, shortly after the wartime rent control in the United Kingdom in September 1939 and Germany's nationwide rent freeze in 1936. The 1939 Order froze the rent *ex post* at the August 1938 level, which may be categorised as a first-generation rent control due to its rigidity under Arnott (1995)'s distinction between 'first-generation' and 'second-generation' rent control regimes, although Arnott's analysis primarily concerns North American and European housing markets. Shortly before the Pearl Harbor attack, the 1941 amendments to the *Land Lease Act* and the *Building Lease Act* sought to give practical effect to wartime rent control by preventing landlords from circumventing rent restrictions through eviction threats and demands for illegal surcharges. The amendments maximised tenure security by introducing automatic renewal mechanisms and the 'just cause' requirement, severely restricting landlords' ability to refuse renewal or terminate leases.

Japan was therefore not an outlier but part of a broader international pattern: many countries introduced or strengthened rent-control and tenant-protection measures in response to war-related inflation, soaring rents, housing shortages, and still-limited general tenant protections (Kholodilin, 2024; Kholodilin et al., 2021). In Japan, however, wartime tenant-protection mechanisms did not remain merely temporary emergency measures. As the March 1945 firebombing significantly reduced Tokyo's housing stock, owing to the combination of densely built urban neighbourhoods and highly flammable wooden housing, the wartime tenant-protection regimes survived the end of the war, as occurred in several European jurisdictions. The postwar *Act on Land and Building Lease* largely inherited the framework established through the wartime amendments, particularly the strong renewal protections and the 'just cause' requirement. This legal framework contributed to the widely recognised landlord perception that 'once leased out, never comes back.' This institutional structure has contributed to persistent risk-averse behaviour among landlords in the Japanese rental housing market.

A.3 Postwar and Post-Bubble Developments

The postwar rental housing system continued to be shaped by strong tenant protections and long-term rental reliance. Although the 2000 amendment introduced fixed-term building leases as an alternative to ordinary leases, the fixed-term system remained relatively uncommon for more than two decades. Ordinary leases, to which automatic renewal and the "just cause" requirement continue to apply, remained the dominant framework.

The prolonged decline in land prices following the burst of Japan's asset-price bubble in the early 1990s further reinforced the relative stability of rental tenancies. During this period, landlords had limited incentives to terminate existing leases for redevelopment or rent increases. At the same time, concerns over negative equity, mortgage availability, and the relatively short economic lifespan of many Japanese housing structures contributed to continued reliance on rental housing among younger and more mobile households.

In this institutional context, Tokyo's rental market differs from housing systems in which renting is primarily treated as a short transitional stage before homeownership. Instead, rental housing constitutes a central and often long-term tenure form. This makes access to rental housing particularly important for social and economic integration, especially for foreign-born residents whose housing opportunities are mediated through private landlords and real-estate agents.

B Survey Administration and Data Quality Procedures

This appendix provides additional details on the implementation of the survey and the data-quality procedures used to construct the final analysis sample.

The survey was administered online between April and May 2025 using the Japanese panel provider *GMO Research*. The target sample consisted of approximately 800 Japanese nationals and 200 foreign-born residents living in the Greater Tokyo area. Foreign-born respondents were oversampled to ensure sufficient statistical power for subgroup analyses. The questionnaire was implemented in Qualtrics and included several quality-control procedures. These included minimum completion-time thresholds, response-completeness checks, location validation, and duplicate-response prevention. Observations were excluded where respondents failed these checks or provided incomplete or internally inconsistent responses.

Ethical approval was obtained from the Department of Land Economy in April 2025, and all participants provided informed consent prior to participation.

C Construction and Validation of Applicant Names

C.1 Selection of Applicant Profiles

Applicant profiles were constructed using *Japanese Immigration Agency* (2024) data to reflect the composition of the 30.7% increase in migrants between 2018 and 2025. The selected profiles correspond to the principal nationality and visa groups represented in Tokyo's foreign resident population. In addition to foreign-born applicant groups, two benchmark profiles were included: a native Japanese applicant and a Western-Japanese applicant (i.e., a foreign-born individual of partial Japanese descent). The latter group was included because prior research suggests that co-ethnicity and perceived cultural proximity may shape evaluations of migrants in Japan (Shipper, 2016).

All applicant characteristics other than names were held constant across profiles in order to isolate perceptions associated with ethnic signals conveyed through names. Applicant identities were standardised as male to reflect prevailing market norms and minimise gender-related variation. This approach is supported by evidence indicating that males may face greater discrimination in rental markets, thereby reducing the need for gender-based stratification (Ahmed and Hammarstedt, 2008; Huang and Bao, 2026). Gender-ambiguous names were avoided to ensure clarity.

C.2 Name Validation Procedure

To ensure that applicant names were recognisable and consistently associated with the intended ethnic categories, a bilingual pre-test was conducted using a sample of 100 respondents recruited through *Prolific*, supplemented by targeted recruitment of Japanese participants.

Respondents were asked to identify the likely ethnic background associated with each name. Names were retained only if they achieved a recognition accuracy threshold of at least 70%, consistent with conventions in correspondence-study research (e.g., Ahmed and Hammarstedt, 2008). Names failing to meet this threshold were excluded from the final survey instrument.

Mean recognisability was higher in English-language responses (94%) than in Japanese-language responses (83%), reflecting linguistic constraints associated with transliteration into Japanese script. While ANOVA results showed no statistically significant differences across names, the model explained more than 60% of the variation in recognition outcomes, indicating meaningful differentiation between applicant categories.

The name validation exercise additionally served as a pilot test for the broader experimental design by assessing whether ethnic signals embedded in names remained sufficiently salient once presented in Japanese script.

C.3 Script Standardisation and Linguistic Design

A key methodological consideration was whether names would remain ethnically distinguishable once translated into Japanese script. To preserve ethnic salience while maintaining comparability across profiles, non-Japanese names were standardised using Katakana characters, which are conventionally used in Japanese for foreign names and loan words. Distinctly Japanese names were presented in Kanji characters. For Western–Japanese profiles, first names were rendered in Katakana and surnames in Kanji to preserve cultural plausibility and reflect mixed ethnic identity.

This approach standardised the presentation of applicant names while enhancing the perceptual salience of ethnic categories that were not Japanese. Prior studies emphasise that Katakana script itself conveys social and ethnic meaning in Japanese contexts (Hanai, 2016). However, loan words written in Katakana do not automatically trigger a specific ethnic association unless supported by cultural familiarity or media exposure (Tamaoka and Miyaoka, 2003). Consequently, respondents' perceptions of names are shaped not only by phonetic structure but also by broader social experience, making Katakana particularly appropriate for studying implicit bias and perceived foreignness.

Particular care was also taken to avoid pronunciation or character ambiguities across East Asian languages. For example, the Chinese surname character 張, commonly pronounced "Zhang" in Mandarin, is pronounced "Chō" in Japanese contexts. To avoid such ambiguities, native Chinese characters were not used directly in the applicant profiles. Instead, names were transliterated into Katakana to ensure that respondents interpreted ethnic signals consistently across profiles.

The final applicant profiles and their corresponding renderings in native script and Katakana are presented in Table 1.

C.4 Translation Procedures

The survey was originally developed in English and translated into Japanese using machine translation, followed by verification by four native speakers. Additional language versions were subsequently developed for Indonesian, Simplified Chinese, English, Hindi, Korean, Burmese, Nepali, Portuguese, and Vietnamese, reflecting the language categories used in the Ministry of Justice Basic Foreign Residents' Survey and covering the largest foreign resident groups in Japan.

The multilingual implementation was designed to maximise accessibility among foreign-born residents while minimising interpretation bias across respondent groups. Japanese was set as the default language option. Active switching to an alternative language version was interpreted as an indicator of lower cultural assimilation and incorporated into the broader analysis of integration-related characteristics.

D Supplemental Questions for Non-Native Japanese Speakers

D.1 Japanese Language Ability

Speaking Ability

Question: How well can you carry a conversation in Japanese?

1. I can carry on a conversation appropriately with anyone, in any situation, on any subject.
2. I can converse fluently and naturally.
3. I can converse as needed in everyday life.
4. I can exchange basic information about family matters.

5. I can use frequently-used greetings and everyday phrases.
6. I cannot carry on a conversation at all.

Reading Ability

Question: How well can you read Japanese?

1. I can read any written material with ease.
2. I can read newspaper articles that are written from certain perspectives.
3. I can read emails that are written using everyday words and phrases.
4. I can read short and easy writings about familiar matters.
5. I can read familiar names and words used in signs and posters.
6. I cannot read well at all.

D.2 Everyday Discrimination Scale (EDS)

Question: In your day-to-day life, how often do any of the following things happen to you?

a) Situations

1. You are treated with less courtesy than other people.
2. You receive poorer service than other people at restaurants or stores.
3. People act as if they are afraid of you.
4. People act as if they think that you are dishonest.

b) Response Scale

- Almost every day
- At least once a week
- A few times a month
- A few times a year
- Less than once a year
- Never

E Supplemental Analyses

E.1 Language Proficiency and Perceived Japanese-Applicant Advantage

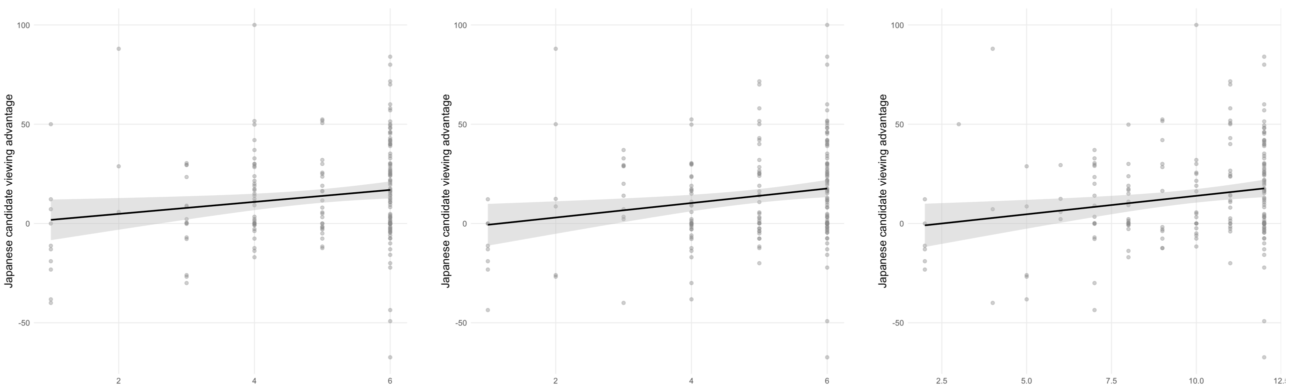
This appendix reports auxiliary analyses examining whether Japanese-language proficiency among foreign-born respondents is associated with perceived relative advantage for the Japanese applicant, complementing results reported in subsubsection 4.6.3. These analyses should be interpreted as suggestive mechanism evidence rather than as direct estimates of GGDD. GGDD refers to differences between majority- and minority-group assessments of the disadvantage facing the same minority group. By contrast, the tests below examine heterogeneity among foreign-born respondents by asking whether respondents with greater Japanese-language proficiency perceive a larger advantage for the Japanese applicant relative to foreign comparison applicants.

The empirical logic is as follows. If Japanese-language access shapes exposure to housing-market information, then foreign-born respondents with greater Japanese proficiency may be more likely to observe Japanese-language rental advertisements, agent communication, public discussion of exclusionary screening, and informal knowledge about search frictions faced by foreign applicants. This may lead them to perceive a larger Japanese-applicant advantage. We examine this possibility using two complementary outcomes.

The first outcome is the gap between the stated viewing likelihood of the Japanese applicant and the average stated viewing likelihood of non-Japanese comparison applicants, excluding the respondent's own profile type and the Western-Japanese profile. Larger values indicate a larger perceived Japanese-applicant advantage. The respondent's own profile type is excluded because it partly captures personal evaluation rather than group-level perception. The Western-Japanese profile is excluded because the main results show that respondents evaluate this profile differently from other non-Japanese profiles, consistent with its partial insider status.

Visual inspection of Figure 6 suggests a threshold pattern: respondents with minimal Japanese proficiency perceive little or no Japanese-applicant advantage, whereas differences among respondents above this lowest-proficiency group appear comparatively small.

Figure 6: Japanese-Language Proficiency and Perceived Japanese-Applicant Advantage



(a) Reading Proficiency (b) Speaking Proficiency (c) Reading and Speaking Proficiency

Notes: The outcome is the stated viewing likelihood of the Japanese applicant minus the average stated viewing likelihood of non-Japanese comparison applicants. Larger values indicate a larger perceived Japanese-applicant advantage. Points show individual foreign-born respondents. Lines show linear fits with 95 percent confidence intervals.

Therefore, rather than assuming a linear relationship, we test whether the graphical pattern is driven by respondents at the lowest level of Japanese proficiency. For each proficiency measure, we compare respondents in the lowest observed category to all other respondents, test whether a categorical specification fits better than a linear specification, and re-estimate the correlation after excluding the lowest category.

Table 10: Lowest Japanese-Proficiency Bracket versus All Other Brackets

A. Lowest proficiency bracket versus all other brackets					
	Lowest	Mean advantage		Difference	
	bracket (N)	All other	Lowest	Lowest – others	p-value
Reading	11	14.89	-7.52	-22.41***	0.005
Speaking	8	14.85	-12.23	-27.07***	0.002
Reading + speaking	6	14.47	-9.03	-23.50**	0.021

B. Fixed Japanese–Chinese applicant gap				
	All foreign-born respondents		Chinese respondents only	
	Correlation (ρ)	p-value	Correlation (ρ)	p-value
Reading	0.180**	0.011	0.220**	0.041
Speaking	0.211***	0.003	0.155	0.151
Reading + speaking	0.207***	0.004	0.201*	0.063

Note: The table compares respondents in the lowest observed Japanese-proficiency bracket with respondents in all higher proficiency brackets. The dependent variable is the Japanese-applicant viewing advantage, defined as the stated viewing likelihood of the Japanese applicant minus the average stated viewing likelihood of non-Japanese comparison applicants, excluding the self-assessment and the Western-Japanese applicant. The difference column reports the estimated difference between the lowest proficiency bracket and all other brackets, calculated as lowest bracket minus all other brackets. Panel B reports Pearson correlation coefficients between Japanese-language proficiency and the fixed Japanese–Chinese applicant gap, defined as the stated viewing likelihood of the Japanese applicant minus the stated viewing likelihood of the Chinese applicant. Larger values indicate a larger perceived advantage for the Japanese applicant. $^{*}p < 0.10$, $**\,p < 0.05$, $***p < 0.01$.

Results reported in Panel A of Table 10 indicate that the association is concentrated among respondents with minimal Japanese proficiency. Respondents in the lowest reading category report a Japanese-applicant advantage that is 22.41 percentage points lower than that of all other foreign-born respondents ($p = 0.005$). The corresponding differences are 27.07 percentage points for speaking proficiency ($p = 0.002$) and 23.50 percentage points for the combined reading-speaking score ($p = 0.021$).

Consistent with this threshold interpretation, treating reading proficiency as a categorical variable fits significantly better than a linear specification ($p = 0.039$). The analogous categorical tests are not statistically significant for speaking ($p = 0.162$) or for the combined proficiency score ($p = 0.274$). After excluding the lowest-proficiency category, the remaining correlations are small and statistically insignificant for reading ($\rho = 0.059$, $p = 0.423$) and speaking ($\rho = 0.091$, $p = 0.214$), and only marginal for the combined score ($\rho = 0.131$, $p = 0.070$). Overall, the auxiliary evidence suggests a threshold pattern at the very lowest levels of Japanese proficiency rather than a general linear association across the full proficiency scale.

The second outcome is a fixed Japanese–Chinese comparison, defined as the stated viewing likelihood of the Japanese applicant minus the stated viewing likelihood of the Chinese applicant.

This comparison complements the aggregate gap in two ways. First, it provides a cleaner comparison because the non-Japanese reference applicant is held constant, avoiding changes in the outcome that may arise from averaging across heterogeneous foreign applicant profiles. Second, it is substantively important. Chinese-origin respondents constitute the largest group among foreign-born participants in our sample, accounting for 44.8% of foreign-born respondents. Chinese nationals also represent 38.0% of Tokyo's registered foreign population, making them the largest foreign nationality group in the city.[32] Moreover, Chinese nationals form the largest group of foreign new arrivals in the migration data used to construct the applicant profiles. For Chinese respondents, the Japanese–Chinese comparison therefore captures an own-group evaluation relative to the Japanese benchmark.

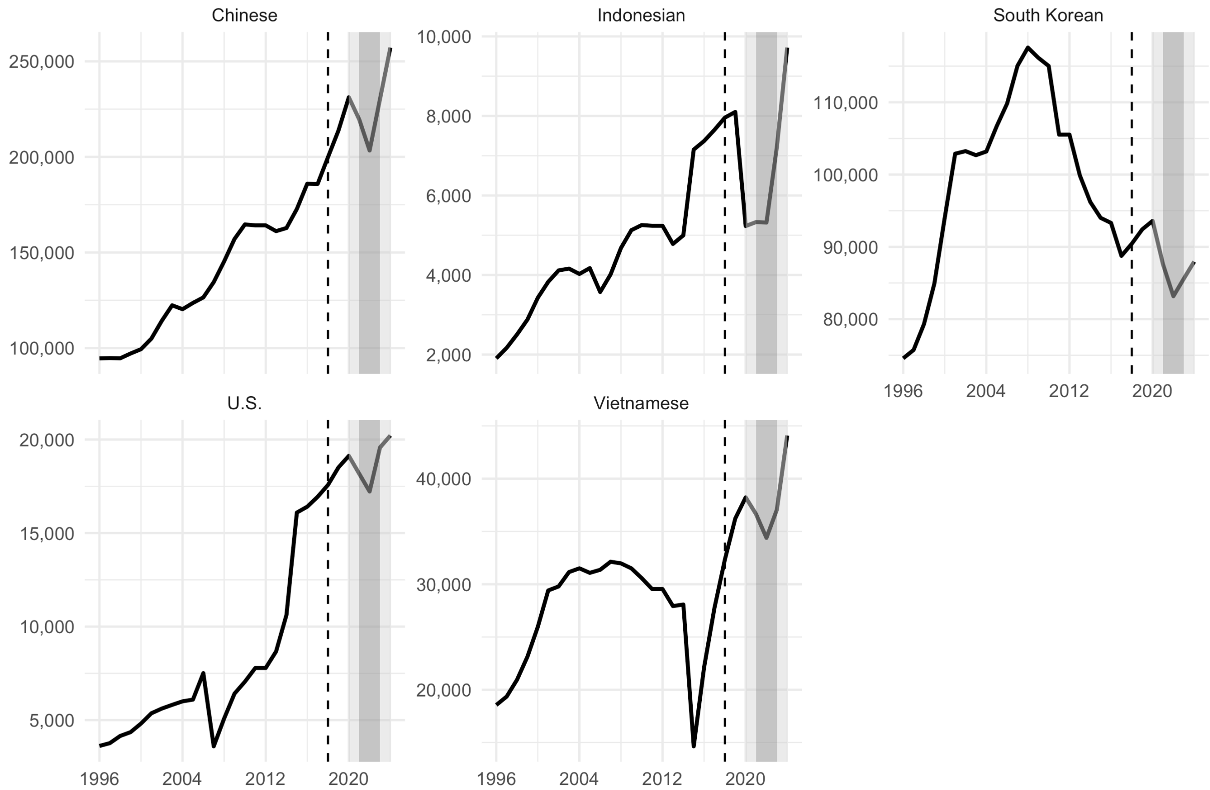
The fixed Japanese–Chinese comparison provides complementary evidence. Among all foreign-born respondents, Japanese-language proficiency is positively associated with the perceived Japanese-applicant advantage over the Chinese applicant. The correlations are positive for reading ($\rho = 0.180$, $p = 0.011$), speaking ($\rho = 0.211$, $p = 0.003$), and the combined reading-speaking score ($\rho = 0.207$, $p = 0.004$). Restricting the sample to Chinese respondents yields substantively similar estimates: $\rho = 0.220$ for reading ($p = 0.041$), $\rho = 0.155$ for speaking ($p = 0.151$), and $\rho = 0.201$ for the combined score ($p = 0.063$). These estimates suggest that Japanese-language proficiency is associated with perceiving a larger Japanese-applicant advantage even when respondents evaluate the Japanese applicant relative to the applicant representing their own nationality group.

These auxiliary results support the interpretation advanced in subsubsection 4.6.3. In the aggregate comparison across non-Japanese applicants, the association is concentrated among respondents with the lowest Japanese proficiency. In the fixed Japanese–Chinese comparison, the association is positive across proficiency measures and remains substantively similar when restricted to Chinese respondents. This pattern strengthens the linguistic-asymmetry interpretation: even when the comparison holds the non-Japanese reference group constant, and even when the analysis is restricted to Chinese respondents, who belong to a large immigrant group likely to have comparatively developed co-ethnic networks and information channels, the substantive association remains similar.

F Additional Figures and Tables

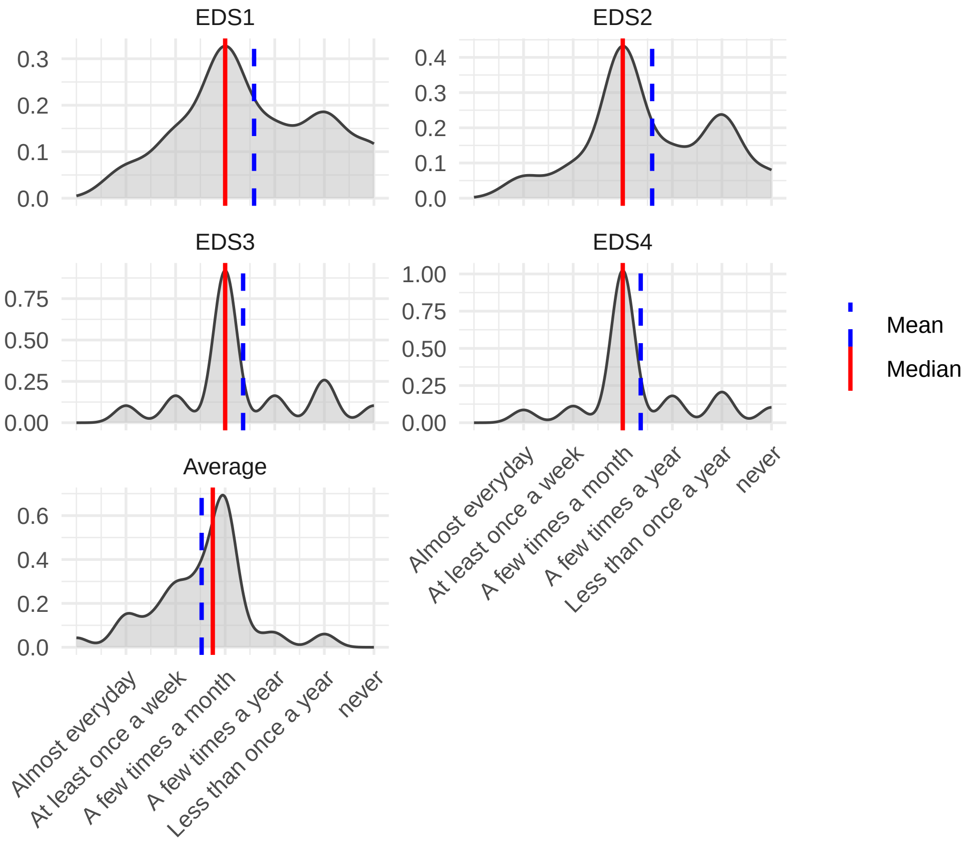
[32] Tokyo Metropolitan Government, *Tokyo Statistical Yearbook 2023*, Table 2-4, "Foreign Residents by District and Nationality," available at `https://www.toukei.metro.tokyo.lg.jp/tnenkan/2023/tn23q3e002.htm`.

Figure 7: Major Nationality Groups among Foreign Residents in Tokyo



Notes: The dashed vertical line marks the 2018 Immigration Control and Refugee Recognition Act reform. Shaded areas indicate the COVID-19 period. The darker shaded region corresponds to years fully affected by pandemic-related restrictions, while lighter shaded regions correspond to partially affected years. Data are from the *Tokyo Statistical Yearbook*.

Figure 8: EDS responses



Notes: The Figure shows densities of each EDS question as well as the density of the composite index achieved as an average across responses.

Table 11: Top Five Visa Groups in Japan

Visa Status	Representative Nationality	Population	Characteristics
Highly Skilled Professional	China (Europe second)	17,912	High skill, high income, high language proficiency
Specified Skilled Worker	Indonesia	44,305	Lower skill, lower income, low language proficiency
Technical Training Intern	Vietnam	44,305	Lower skill, lower income
Permanent Resident	China	336,086	Higher skill, high language proficiency
Special Permanent Resident	South Korea	250,881	Cultural ties, high integration

Note: Data obtained from the Japanese Immigration Agency (2024).

Table 12: Summary Statistics

Variable	Mean / Share (%)	P25	Median	P75	SD
All respondents					
Age	48.10	39.00	50.00	57.00	12.60
Income (brackets)	6.05	2.00	6.00	9.00	3.99
Japanese	80.50%				
Female	40.60%				
Owner-occupier	52.80%				
Degree	58.00%				
Full-time employed	47.20%				
Non-Japanese residents					
Age	45.10	36.00	45.00	53.00	13.40
Income (brackets)	6.48	3.00	7.00	10.00	4.01
Female	51.20%				
Owner-occupier	52.70%				
Degree	70.90%				
Full-time employed	52.70%				
Length of stay	5.27	5.00	5.00	7.00	1.55
Speaking proficiency	4.94	4.00	5.00	6.00	1.36
Reading proficiency	4.94	4.00	6.00	6.00	1.43
Permanent visa	61.10%				

Notes: The Table reports summary statistics on all variables used. Numbers are computed for the full sample as well as the subsample of non-Japanese participants.

Table 13: Highest- and Lowest-Ranked Non-Japanese Applicants by Respondent Nationality

Respondent nationality	Highest-ranked applicant	Lowest-ranked applicant
Japan	Western-Japanese	Indonesian
China	Western-Japanese	Chinese
Taiwan	Western-Japanese	Vietnamese
South Korea	Western-Japanese	Vietnamese
Vietnam	Western-Japanese	South Korean

Notes: This Table complements the aggregate rankings shown in Figure 5 by reporting the highest- and lowest-ranked non-Japanese applicants within selected respondent-nationality groups. Nationality-specific rankings for smaller respondent groups should be interpreted descriptively due to limited sample sizes.

G Experimental Design Document and Participant Instructions

This appendix consolidates the design-document and participant-instruction material. Where the manuscript's other appendices (Appendix A–Appendix D) already cover the relevant material, those appendices are cross-referenced rather than duplicated.

G.1 Experimental design overview

Setting and target population. Perceived ethnic discrimination in the private rental housing market of the Greater Tokyo Metropolitan Area, among adult residents who are either Japanese citizens or foreign-born residents legally resident at the time of fieldwork.

Recruitment. The survey was administered through the *GMO Research* online panel via the *Qualtrics* platform. GMO Research was selected over Amazon Mechanical Turk (which has limited Japanese penetration) and operates a foreign-resident sub-panel suitable for the oversampled foreign-born quota. Detail: Appendix B.

Sampling frame and quotas. Quota sampling: target $\approx$ 800 native Japanese and $\approx$ 200 foreign-born respondents resident in Greater Tokyo. Foreign-born respondents were intentionally over-sampled relative to their population share ($\approx$ 5% of Tokyo residents) to ensure sufficient statistical power for the EDS-interaction analyses (subsubsection 4.6.2) as well as the personal–group and group–group comparisons (subsubsection 4.5.1 and subsubsection 4.5.2). Realised sample summary statistics are reported in Table 12.

Eligibility. Respondents had to (i) be aged 18 or above; (ii) reside in Greater Tokyo at the time of the survey (validated by geolocation and a screening question); and (iii) not have participated in a related pilot. Foreign-born respondents were further required to self-identify as non-Japanese citizens.

Fieldwork window. Data collection took place in April–May 2025.

Informed consent. A written briefing was presented before participants entered the substantive questionnaire (text reproduced in subsection G.4 below). Participants gave informed consent by progressing past the briefing screen; withdrawal was permitted at any time without penalty.

G.2 Treatment design and randomisation

The experimental treatment varies the ethnic signal carried by the applicant's name while holding all other applicant characteristics constant. The seven applicant profiles are documented in Table 1 and constructed using the procedure described in Appendix C. Each respondent evaluates the perceived viewing-invitation likelihood for each of the seven profiles in turn, evaluated against the same standardised property listing sourced from *LIFULL Homes*.

Within-subjects evaluation. The within-subjects design allows identification of relative ethnic-signal effects free of between-respondent compositional differences.

Personal-prospect question. After evaluating the chances of the seven applicants, respondents are asked to assess the probability that *they personally* would be invited to a viewing for the same property. This approach together with respondents' stated own origin enables us to measure the personal-group and group-group discrimination discrepancy analyses (subsubsection 4.5.1, subsubsection 4.5.2).

Treatment-order randomisation. The order in which the seven applicant profiles were presented was randomised across respondents using Qualtrics' randomisation logic, neutralising order-of-presentation effects. The per-respondent presentation order is recorded in the raw Qualtrics export and is reproducible from the deposited dataset.

Sample-size justification. The quota targets were set ex ante on the basis of (i) the GMO Research foreign-resident sub-panel reach estimate of 1,200–1,400 reachable foreign-born respondents; (ii) detectable-effect-size considerations for the planned subgroup analyses; and (iii) budget constraints under the CULS and Department of Land Economy funding award. No formal pre-registered power calculation was undertaken; the absence of formal pre-registration is acknowledged in the *Declarations* (Pre-registration disclosure) and in the cover letter.

G.3 Data quality procedures

See Appendix B for the full data-quality protocol (minimum completion-time thresholds, response-completeness and consistency checks, geolocation and device-metadata validation, duplicate-response prevention, pilot phase, multilingual default-language tracking).

G.4 Participant instructions: Japanese and English

This subsection provides the participant-facing instructions in Japanese (the language in which the survey was fielded by default) and the English translation used for instrument development.

Welcome and informed-consent briefing (QID1)

English (fielded version).

> The purpose of this survey is to collect data for an undergraduate dissertation project at the Department of Land Economy, University of Cambridge. The survey should take approximately 5 minutes, and your responses will remain anonymous. The collected data will be used for academic research purposes only, and will not be used for any commercial purposes.
>
> You can only take this survey once. You cannot go back to change your previous answers, so please answer the questions to the best of your ability. You can discontinue the survey at any point; however, participants will be compensated only upon completion of the survey. By checking the box below, you consent to our using the data for academic research. Thank you kindly for your time.
>
> I have read the introductory information and give my consent to the use of this data for academic research. *[Response options: Yes / No.]*

Japanese (fielded version).

> 本調査の目的は、「都市部の居住分離に対する経済的および文化的影響: 住宅市場からの証拠」というプロジェクトのデータ収集です。調査には約 5 分かかり、回答は匿名のまま収集されます。データは学術研究目的にのみ使用され、最長 5 年後に削除されます。
>
> 本調査は 1 回のみ受けることができます。以前の回答を変更することはできませんので、できる限り正確に質問に回答してください。調査はいつでも中止することができますが、参加者への報酬は調査完了時にのみ支払われます。下のボックスにチェックを入れることで、学術研究のためにデータを使用することに同意したものとみなされます。ご協力ありがとうございます。

概要を読み、データを学術研究に使用することに同意します。*[*回答選択肢*:* はい / いいえ*]*

Section 1: socio-economic characteristics

The questionnaire transitions directly from the consent screen to the demographic block; there is no separate Section 1 instruction screen. Demographic items follow the wording of the Japanese Ministry of Justice *Basic Foreign Residents' Survey* and the *2020 National Census.* The first item is reproduced in both languages as an example.

Example item (QID2) — English.

What is your nationality of birth? *[Response options: Japan; China; South Korea; Vietnam; Philippines; Brazil; Nepal; Indonesia; Taiwan; USA; Thailand; Other (please specify).]*

Example item (QID2) — Japanese.

あなたの生まれた国はどこですか? *[*回答選択肢*:* 日本 / 中国 / 韓国 / ベトナム / フィリピン / ブラジル / ネパール / インドネシア / 台湾 / アメリカ / タイ / その他 *(*具体的にご記入ください*)]*

Section 2: viewing-likelihood evaluation

Each respondent evaluates seven applicants in random order (one per Qualtrics screen), all against the same standardised *LIFULL Homes* property listing (Figure 2). The two examples below show (a) the wording for the Japanese-baseline applicant (Taro Yamamoto, QID7) and (b) the personal-prospect (own) question (QID14) on which the personal–group discrimination discrepancy analysis in subsubsection 4.5.1 depends. The remaining six applicant items (QID8–QID13) follow the QID7 template with only the applicant name changed.

QID7 (Taro Yamamoto) — English.

Taro Yamamoto has just moved to Tokyo; he wants to rent the above property. He contacted the landlord and asked to view the property. What do you think is the chance of him being invited to a viewing? *[Response on an 11-point scale, 0%–100% in 10-point increments.]*

QID7 (Taro Yamamoto) — Japanese.

山本太郎さんは東京に引っ越してきたばかりで、上記の物件を借りたいと考えています。彼は家主に連絡して物件を内覧したいと申し出ました。彼が内覧に招待される可能性はどれくらいだと思いますか? *[11* 段階尺度*, 0%* から *100%* まで *10%* 刻みで回答*]*

QID14 (personal-prospect) — English.

If you want to rent the above property, and you have just contacted the landlord and asked to view the property, what do you think is the chance of you being invited to a viewing? *[Response on the same 11-point scale.]*

QID14 (personal-prospect) — Japanese.

> あなたが上記の物件を借りたいと思って、家主に連絡して物件を内覧したいと頼んだとします。内覧に招待される可能性はどれくらいだと思いますか? *[同じ 11 段階尺度で回答]*

Section 3: Everyday Discrimination Scale (foreign-born respondents only)

Adapted from Williams et al. (1997). The intro and four items are reproduced verbatim below; the response scale is shared across the four items (*Almost every day* / *At least once a week* / *A few times a month* / *A few times a year* / *Less than once a year* / *Never*). The English wording is also reprinted in Appendix D.

QID33 intro — English.

> In your day-to-day life, how often do any of the following things happen to you:
>
> 1. You are treated with less courtesy than other people.
> 2. You receive poorer services than other people at restaurants or stores.
> 3. People act as if they are afraid of you.
> 4. People act as if they think that you are dishonest.

QID33 intro — Japanese.

> 日常生活の中で、以下のような事柄がどのくらいの頻度で起こりますか?
>
> 1. 他の人よりも丁寧に扱われない。
> 2. レストランや店で他の人よりも質の悪いサービスを受ける。
> 3. 人々があなたを恐れているかのように行動する。
> 4. 人々があなたを不誠実だと思っているかのように行動する。

Declarations

Funding

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Ethics approval

This study was approved on 5 March 2025 by the Departmental Research Committee of the Department of Land Economy, University of Cambridge (reference DC05032025). All four co-authors are affiliated with the University of Cambridge, and a single Cambridge ethics approval covers the survey administered to native Japanese and foreign-born respondents in Greater Tokyo via the GMO Research panel in April–May 2025. Participants provided informed consent prior to participation.

Conflict of interest

The authors declare that they have no relevant financial or non-financial interests to disclose.

Author contributions

Jessica Alder: original research design, lead survey-instrument design, primary author of the empirical analyses and original dissertation work from which this paper was developed. **Sofie R. Waltl**: methodological lead, empirical analyses, supervisory input, manuscript revision, and overall framing. **Kate Konatsu Nishigai**: Japanese-language validation, legal expert, translation oversight, and manuscript revision. **Helen X. H. Bao**: senior author and corresponding author; conceptualization, investigation, project supervision, funding acquisition, manuscript revision, and submission.

Pre-registration disclosure

The study was not formally pre-registered with the AEA RCT Registry or an equivalent public registry. The empirical hypotheses tested in the paper were pre-specified in the analysis-script structure prior to the analysis, but no public pre-registration document exists.

Data availability statement

The data underlying this article were collected through an online survey administered via the GMO Research panel (Tokyo, Japan) and processed using Qualtrics. The cleaned, de-identified individual-level dataset, the survey instrument (Japanese and English versions), and all analysis code (R) that reproduce the results in the manuscript and the appendix will be deposited in a replication repository upon acceptance of this manuscript. De-identification entails removal of all GMO Research panel-member identifiers and panel-onboarding metadata; suppression of direct and indirect personally-identifying variables; and coarsening of small-cell demographic combinations where residual re-identification risk would otherwise be material. The complete archive is held by the corresponding author and may be shared on reasonable request pending the public deposit.